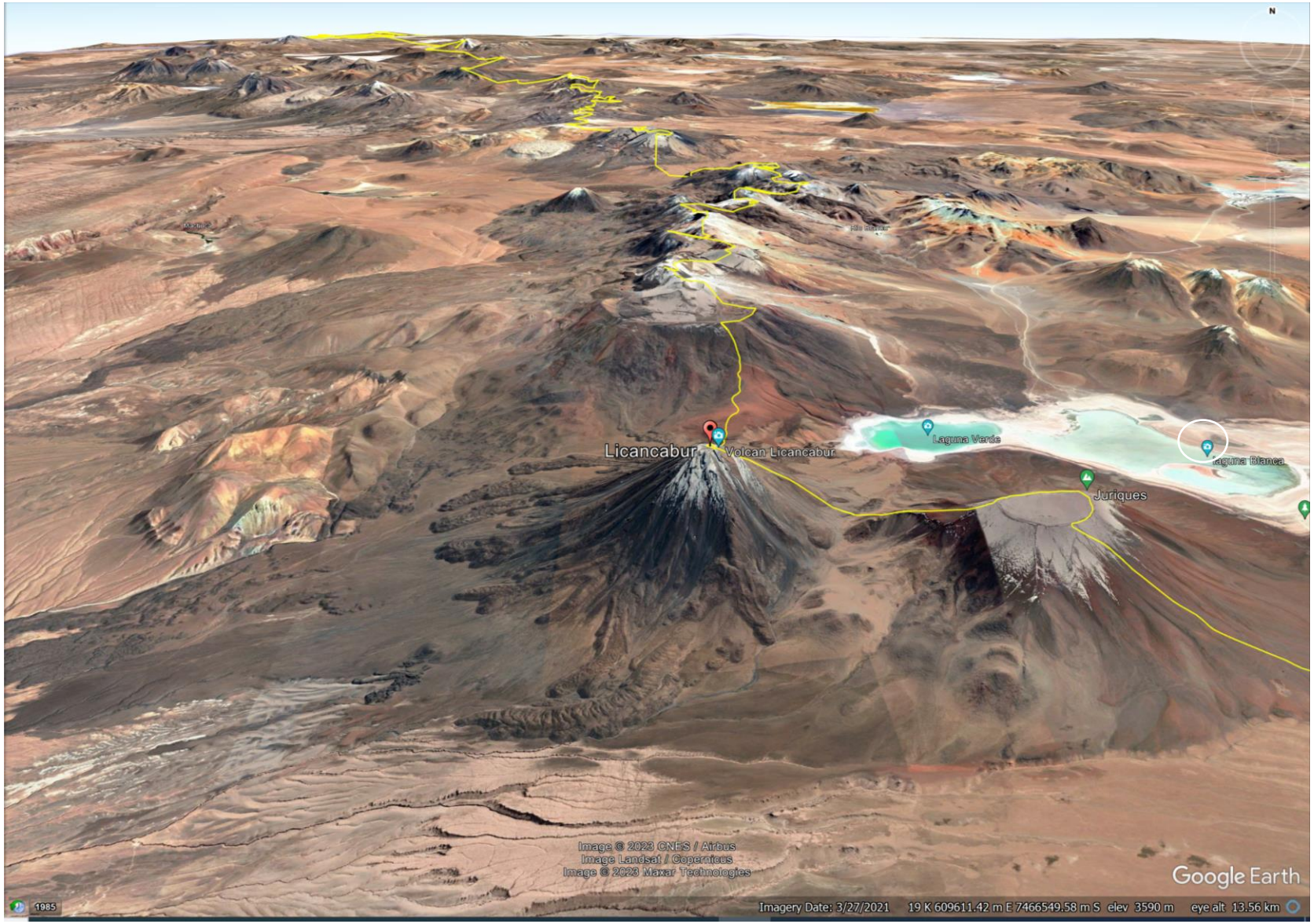


LICANCABUR LAHARZ Exercise





Atacama desert with almost no precipitation all year

Licancabur

6km

12km

15km

Image © 2023 CNES / Airbus
Image © 2023 Maxar Technologies

Google Earth

Imagery Date: 11/9/2021 22°51'18.30" S 67°56'11.54" W elev 3693 m eye alt 19.11 km



Licancabur

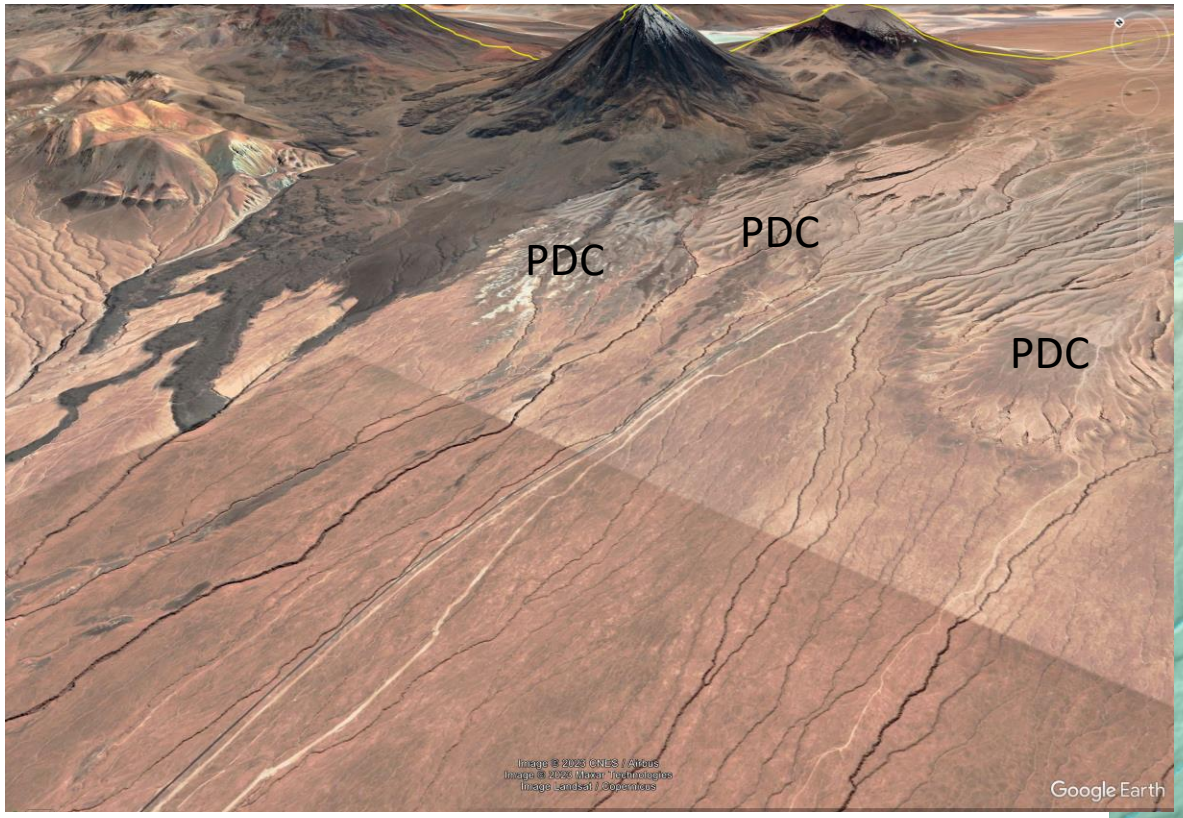
Licancabur

CHILE

15km

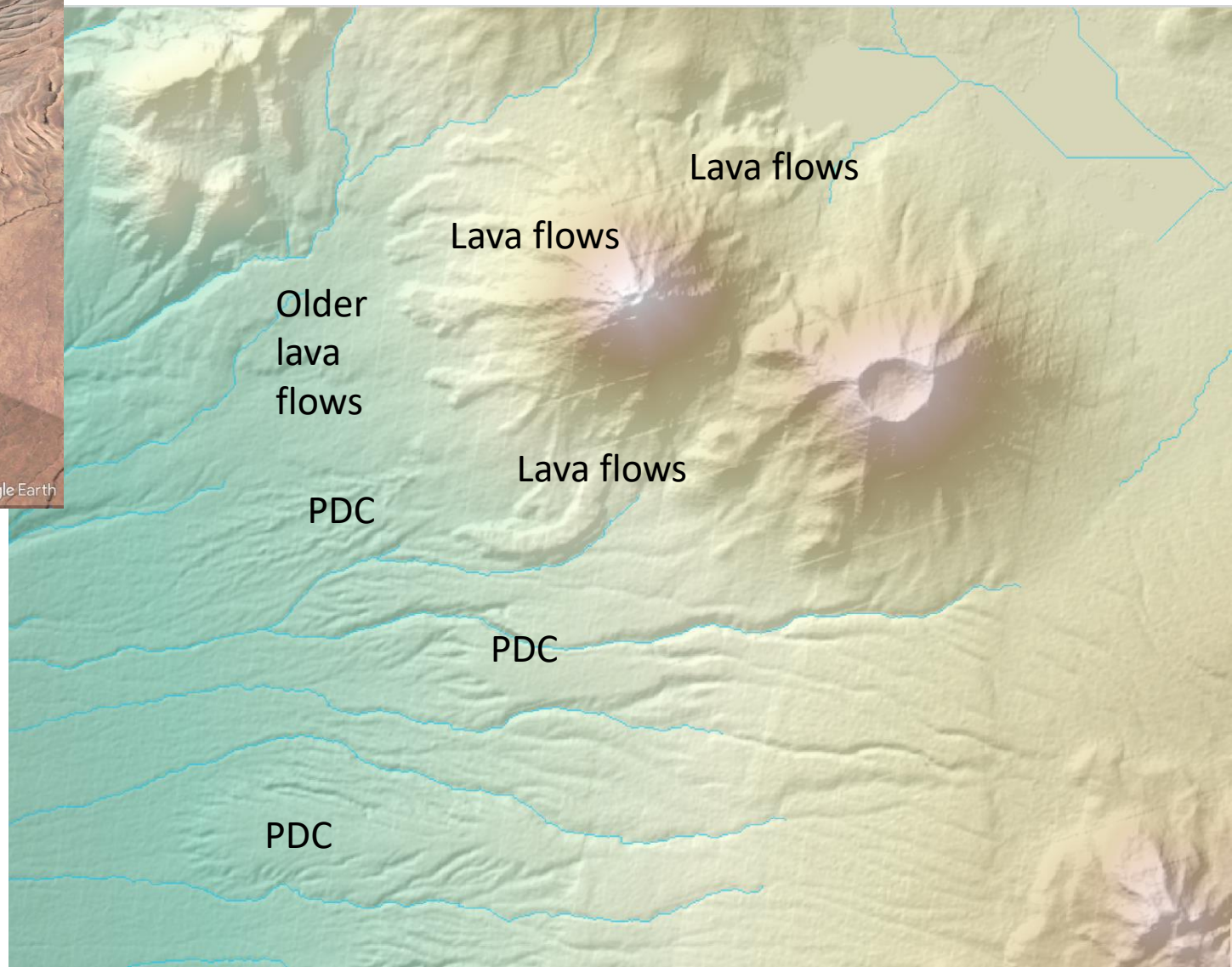
Google Earth

Imagery Date: 11/9/2021 22°51'18.30" S 67°56'11.54" W elev 3693 m eye alt 19.11 km



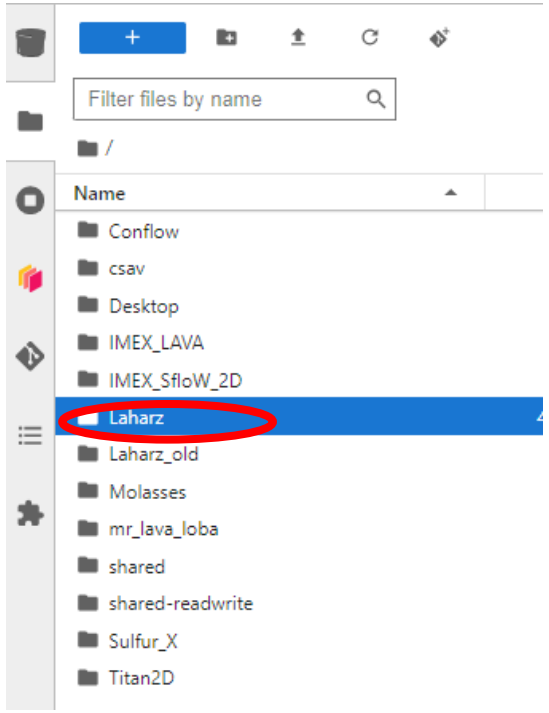
Google Earth and satellite imagery

Geomorphology seen in DEM and shaded-relief map

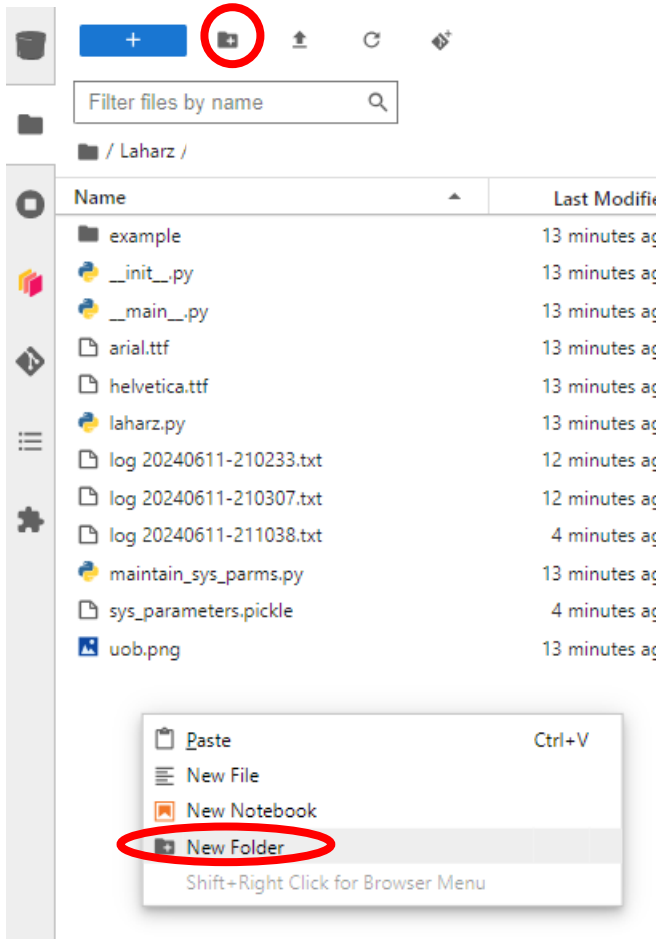


PDCs = Pyroclastic density currents

Let’s make a folder to work in!

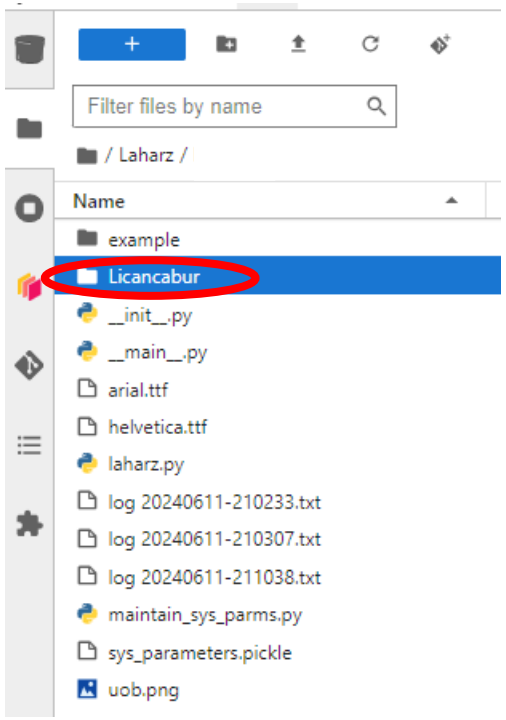


Double click on “Laharz”

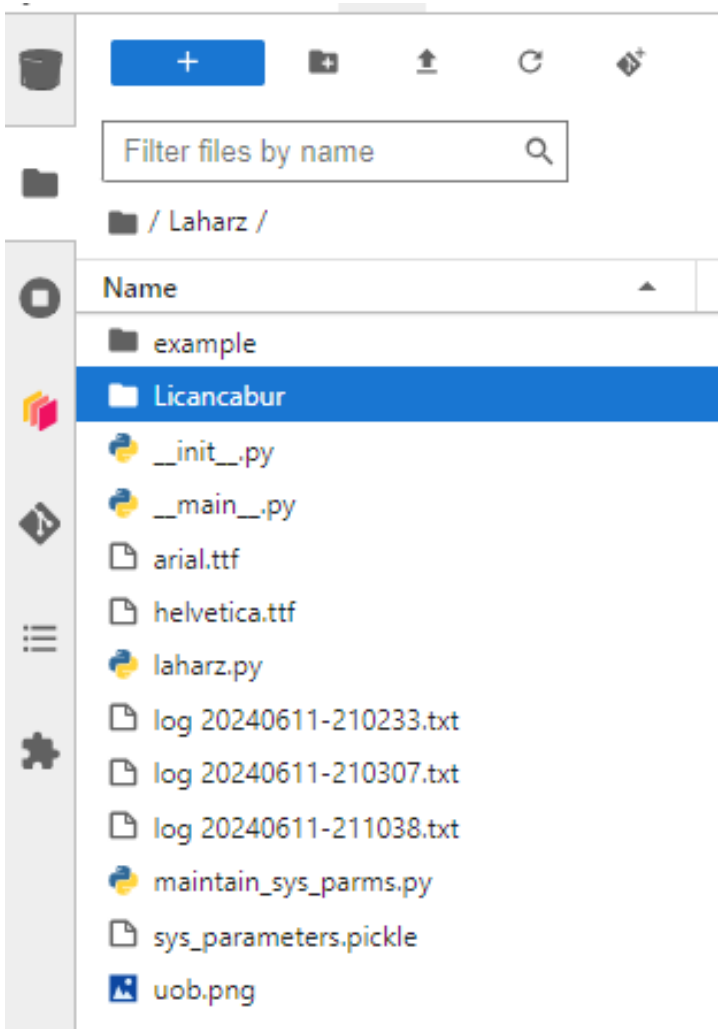


Right click in a blank space and choose “New Folder”

Or click the “New folder“ icon at the top



Name the new folder “Licancabur”



When you work on your projects, you'll want to make a new folder here too /Laharz/new_volcano_name (so if my project is on Mt. St. Helens I might make a folder called /Laharz/Helens)

Download a DEM

- Run the “DEM_download.ipynb” notebook again to download a DEM using these coordinates:

Specify parameters

In the box below, edit the values to specify the extent of the DEM and the output filetype.

[10]:
```
#Edit the latitude and longitude of the DEM extent in decimal degrees
north =  -22.73174

south = -23.00095

west = -68.27182

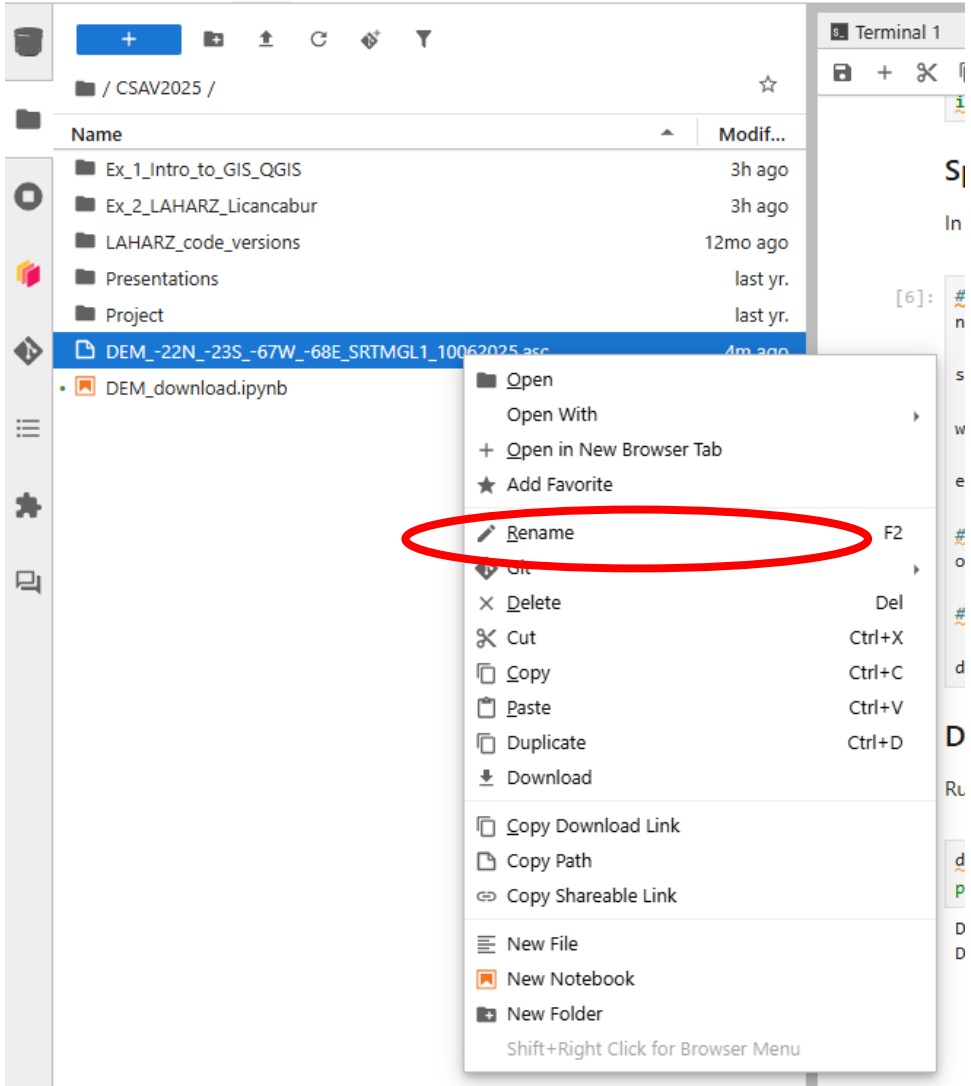
east =  -67.76064

#Select a DEM format: Input "ascii" for ASCII format (.asc) and "tiff" for geotiff format
outputFormat = "ascii"

#Choose the Open Topography data type you want to download

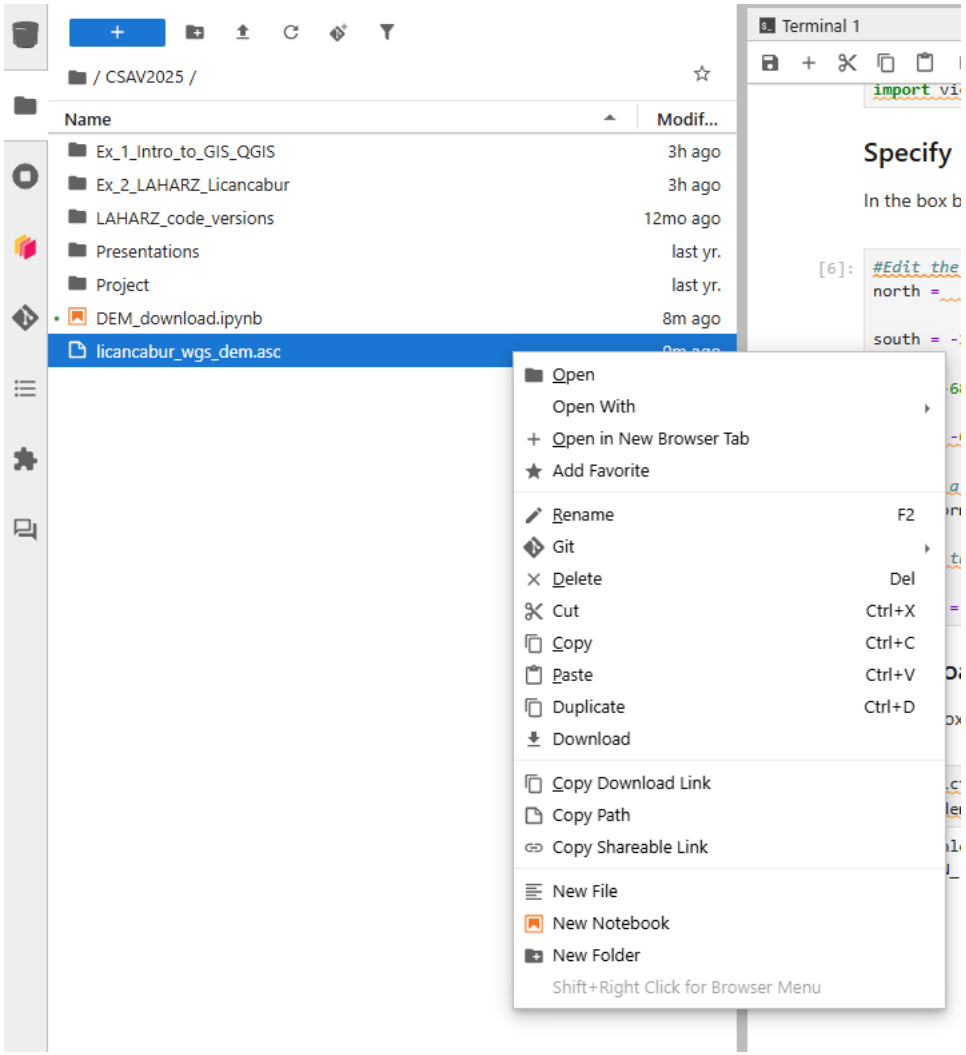
dataset = "SRTMGL1"
```

Rename the DEM

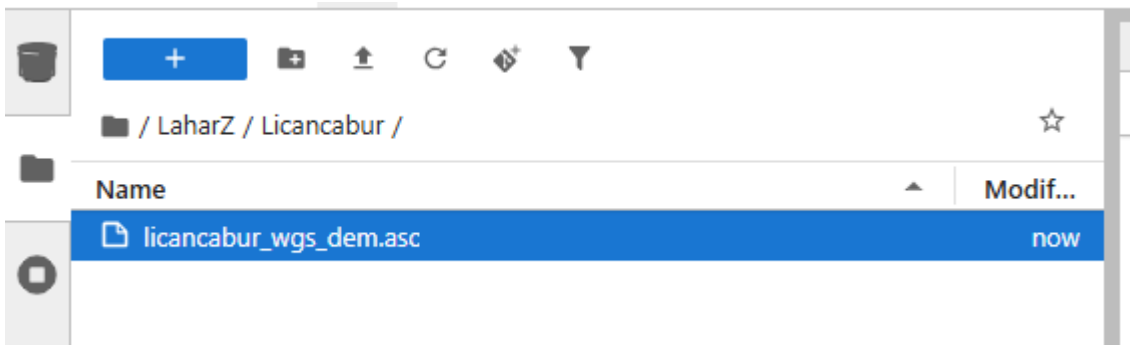


- The downloaded DEM should now be in your CSAV2025 folder
- Rename it by right clicking and choosing “Rename”
- Name it “licancabur_wgs_dem.asc”

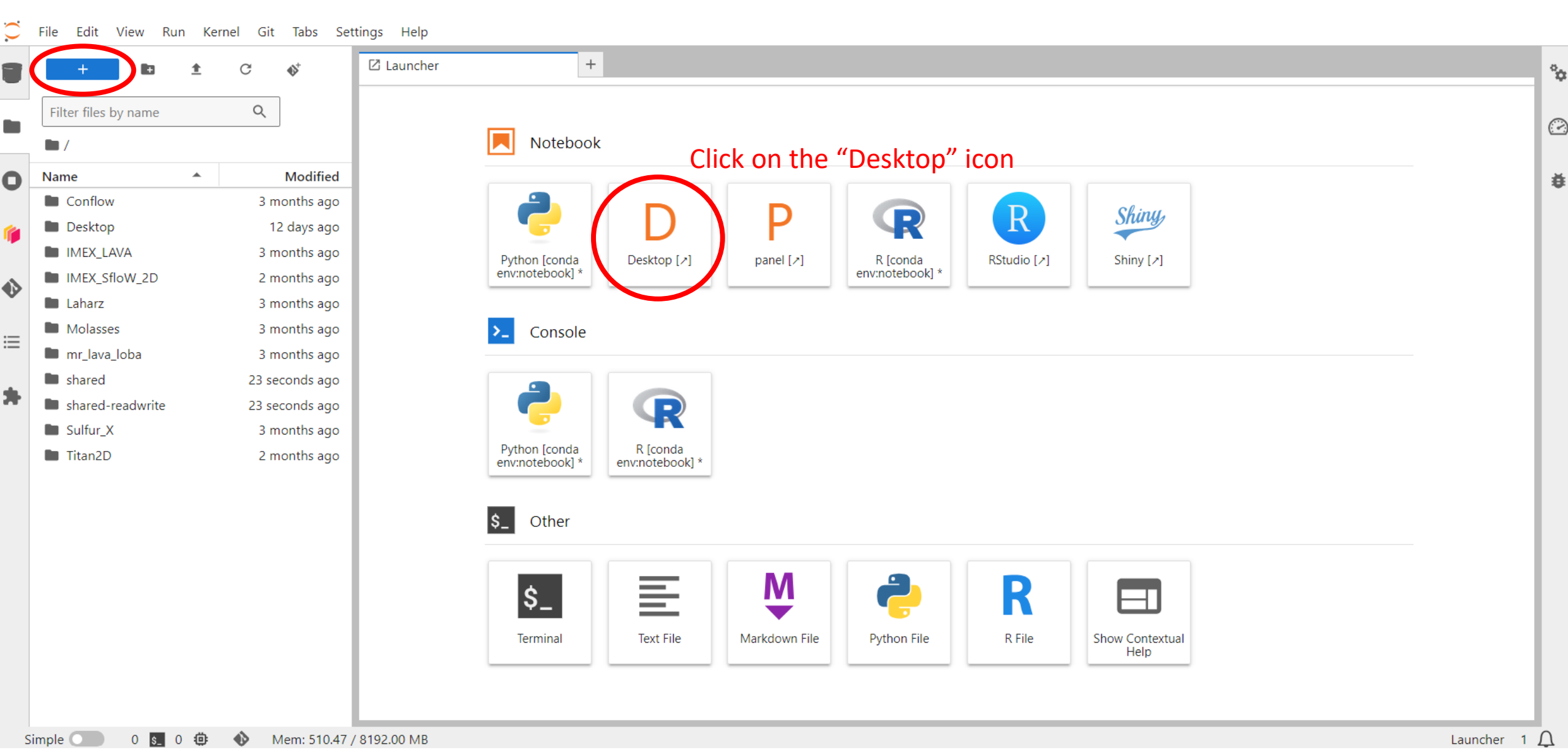
Move the downloaded Licancabur DEM to the new folder



Right click on your DEM and choose “Copy”

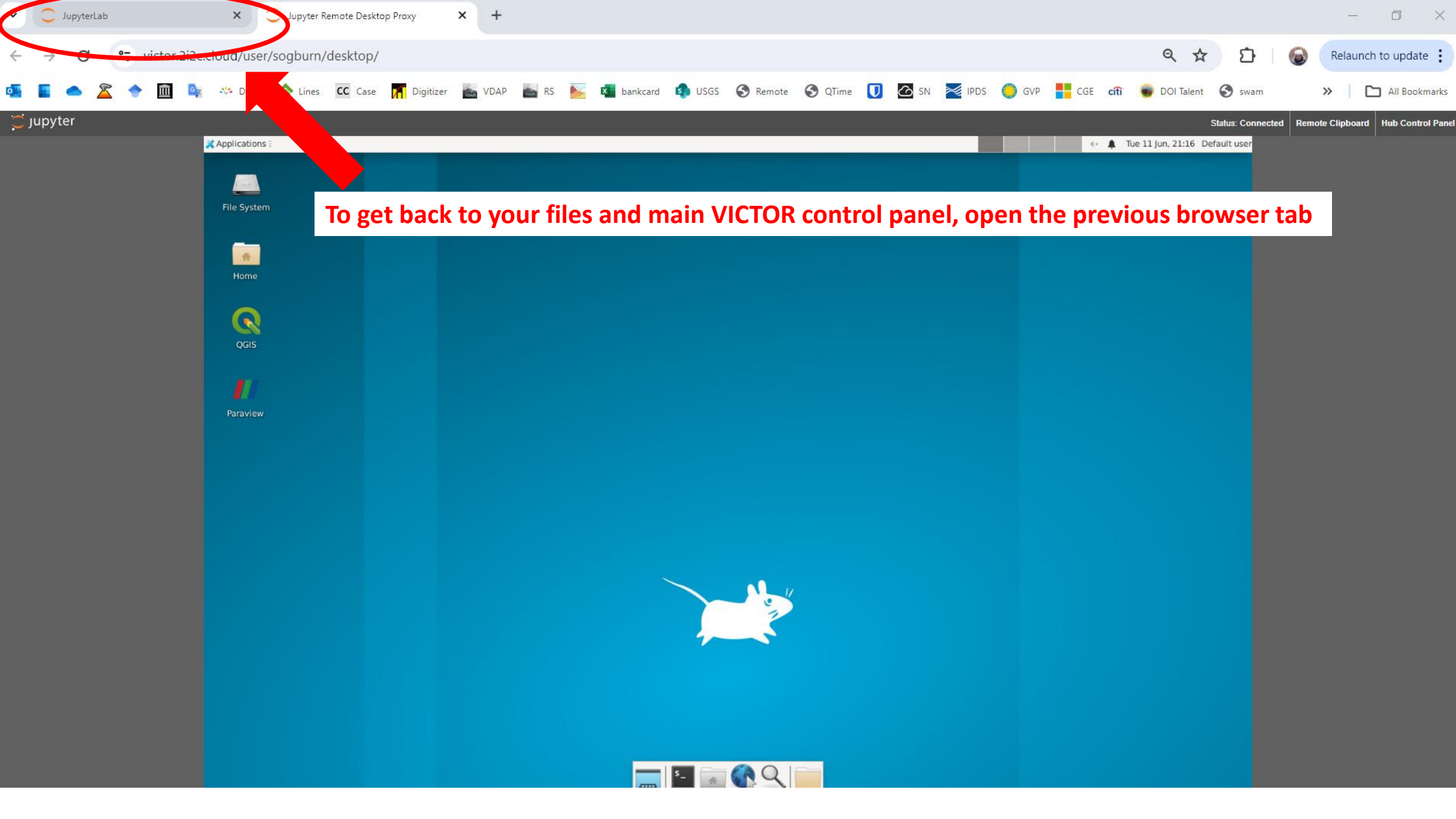


Navigate to LaharZ/Licancabur, right click, and choose “Paste”

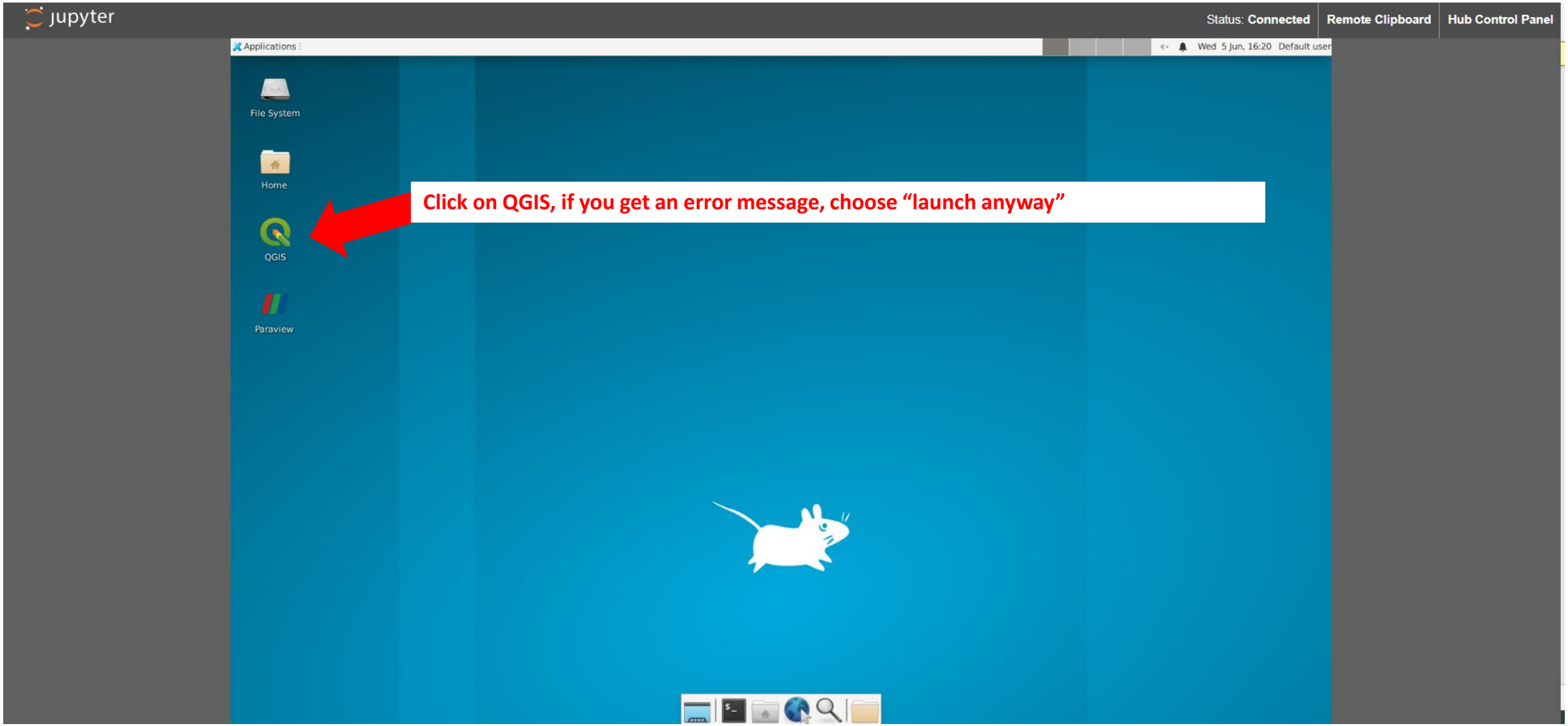


Click on the “Desktop” icon

Tip: If you lose your “Launcher” screen, click the “plus” icon at the top left for “New Launcher”


JupyterLab
Jupyter Remote Desktop Proxy
victor.2i2c.cloud/user/sogburn/desktop/
Relaunch to update
Lines
Case
Digitizer
VDAP
RS
bankcard
USGS
Remote
QTime
SN
IPDS
GVP
CGE
DOI Talent
swam
All Bookmarks
jupyter
Status: Connected
Remote Clipboard
Hub Control Panel
Applications
Tue 11 Jun, 21:16
Default user
File System
Home
QGIS
Paraview
To get back to your files and main VICTOR control panel, open the previous browser tab

Let’s make a filled DEM.



Applications | LaharZ 2.1.1 | Untitled Project — QGIS | Terminal | Tue 11 Jun, 23:04 Default use

Untitled Project — QGIS

Project Edit View Layer Settings Plugins Vector Raster Database Web Mesh Processing Help

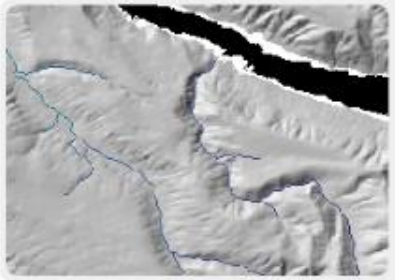
Click on “New Project”

Browser

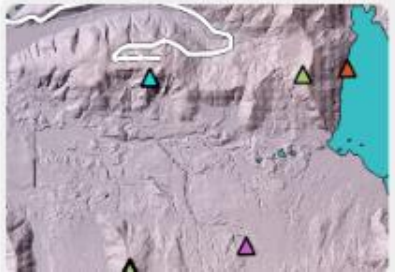
Favorites
Spatial Bookmarks
Home
/
GeoPackage
SpatiaLite
PostgreSQL
MS SQL Server
WMS/WMTS
Vector Tiles
XYZ Tiles
WCS
WFS / OGC API - Features
ArcGIS REST Servers



ol
yan/Laharz/laharz/Licancabur/large_vol.qgz
EPSG:4326 (EPSG:4326 - WGS 84)

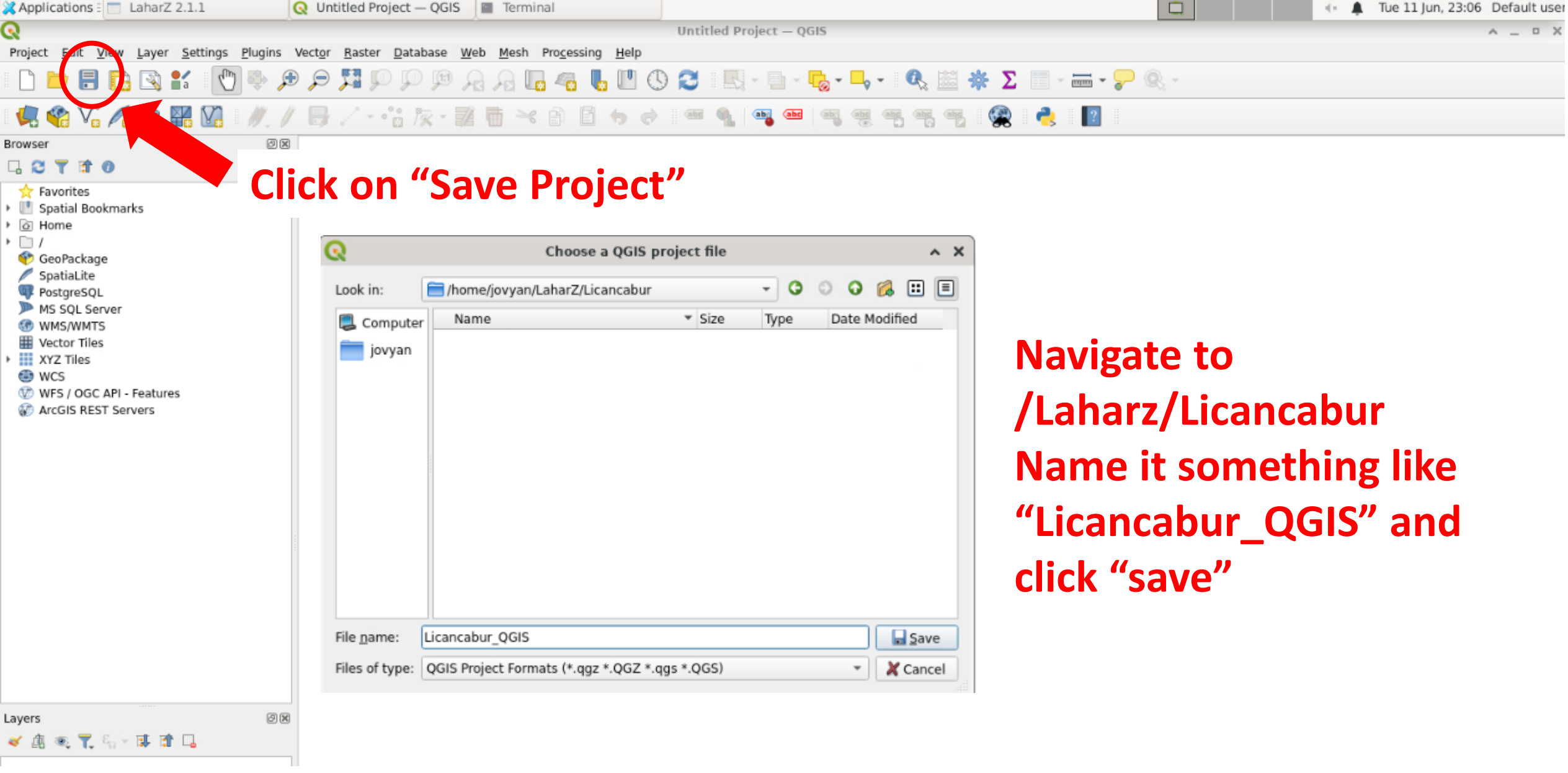


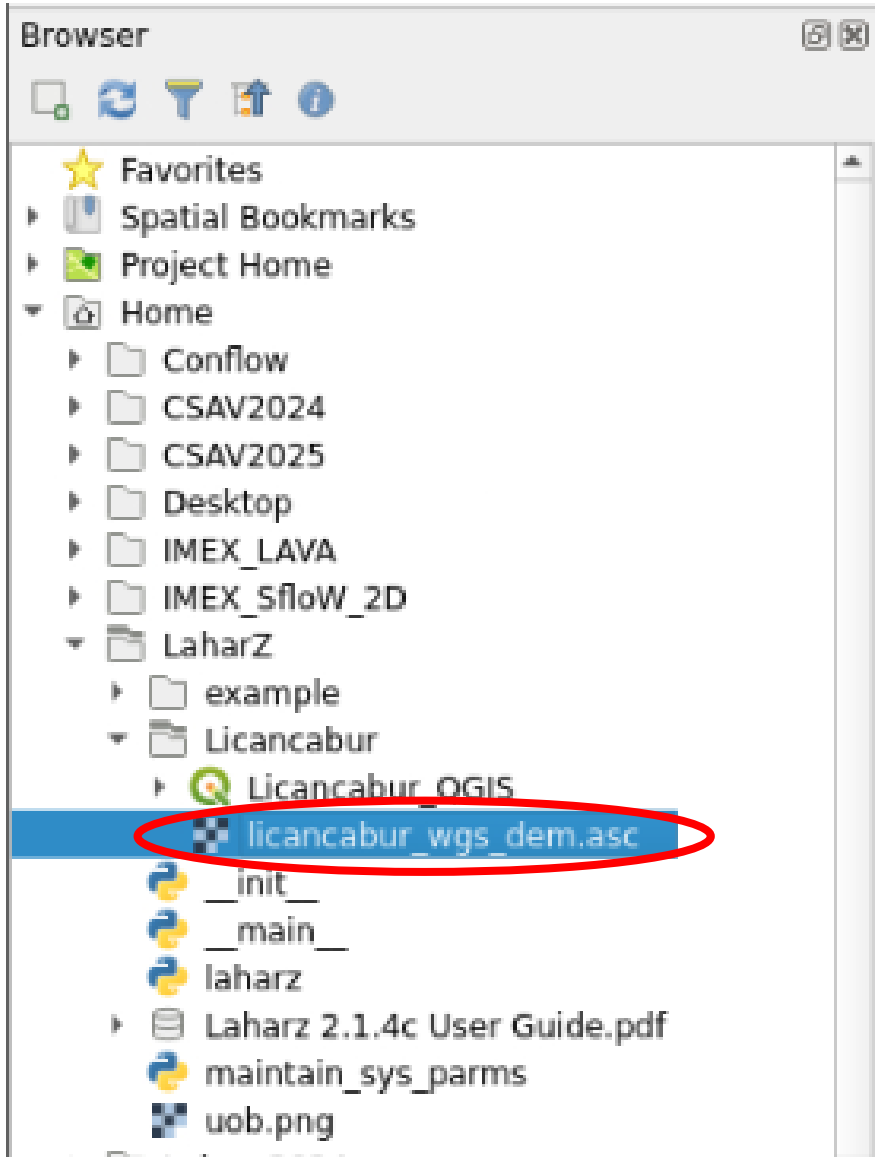
wgs_outputs
/home/jovyan/Laharz/laharz/rainier/wgs_outputs.qgz
EPSG:4326 (EPSG:4326 - WGS 84)



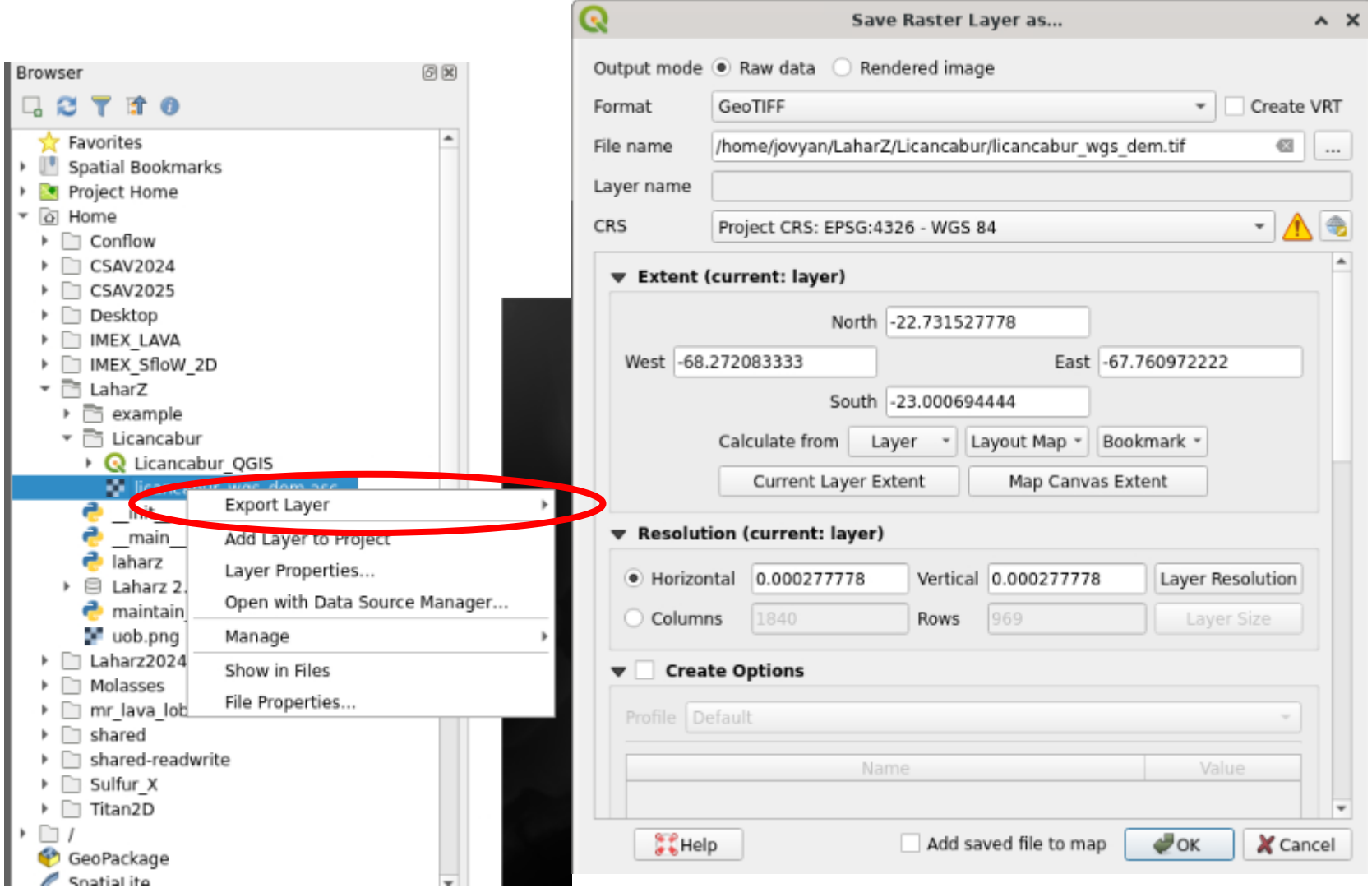
MSH_GIS
/home/jovyan/Laharz/GIS_exercise/MSH_GIS.qgz
EPSG:26710 (EPSG:26710 - NAD27 / UTM zone 10N)

Project Templates

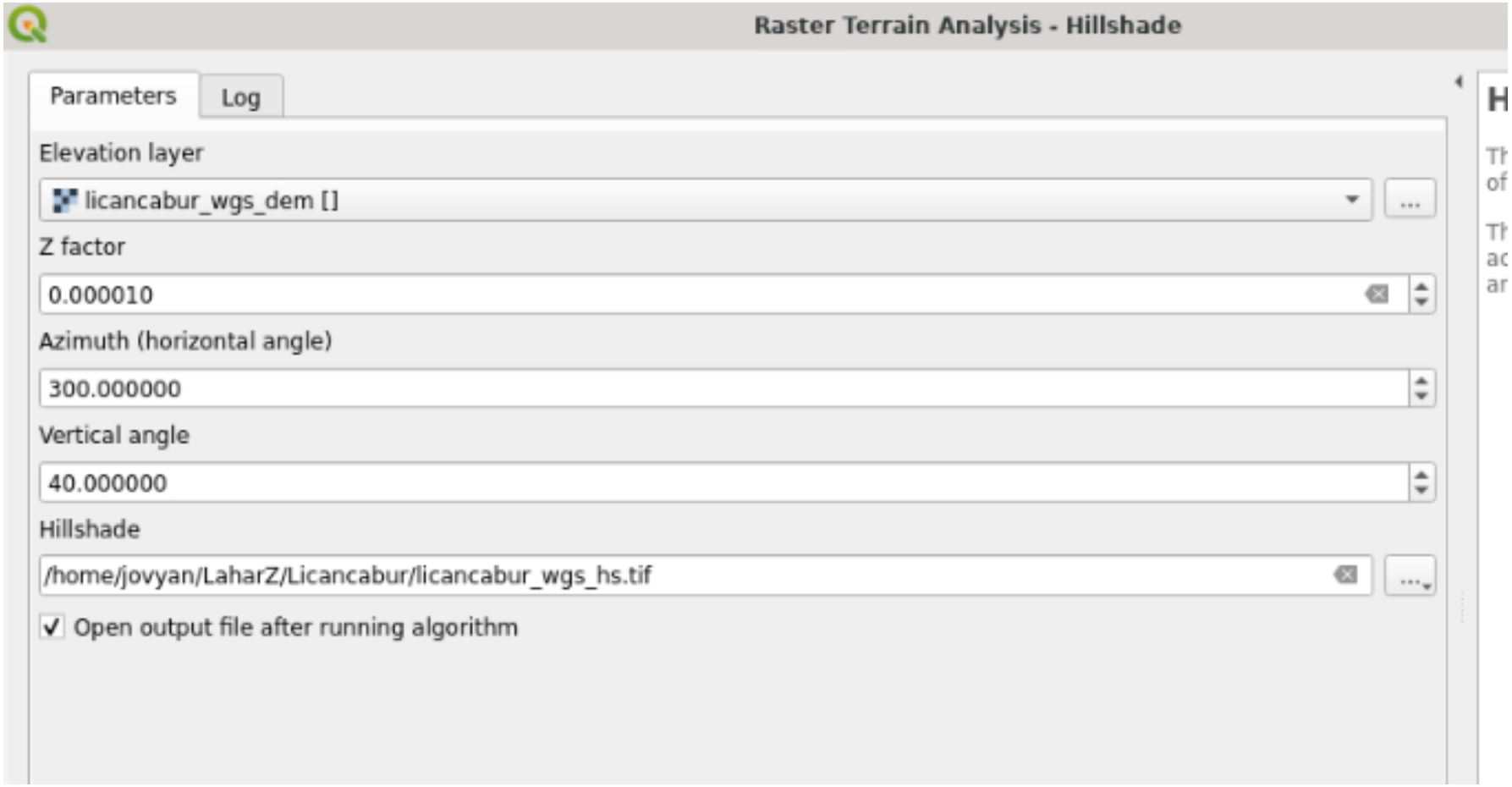

Applications
LaharZ 2.1.1
Untitled Project — QGIS
Terminal
Tue 11 Jun, 23:06
Default user
Untitled Project — QGIS
Project Edit View Layer Settings Plugins Vector Raster Database Web Mesh Processing Help
Browser
Favorites
Spatial Bookmarks
Home
/
GeoPackage
SpatiaLite
PostgreSQL
MS SQL Server
WMS/WMTS
Vector Tiles
XYZ Tiles
WCS
WFS / OGC API - Features
ArcGIS REST Servers
Layers
Click on “Save Project”
Choose a QGIS project file
Look in:
/home/jovyan/LaharZ/Licancabur
Computer
jovyan
Name
Size
Type
Date Modified
File name:
Licancabur_QGIS
Save
Files of type:
QGIS Project Formats (*.qgz *.QGZ *.qgs *.QGS)
Cancel
Navigate to
/Laharz/Licancabur
Name it something like
“Licancabur_QGIS” and
click “save”



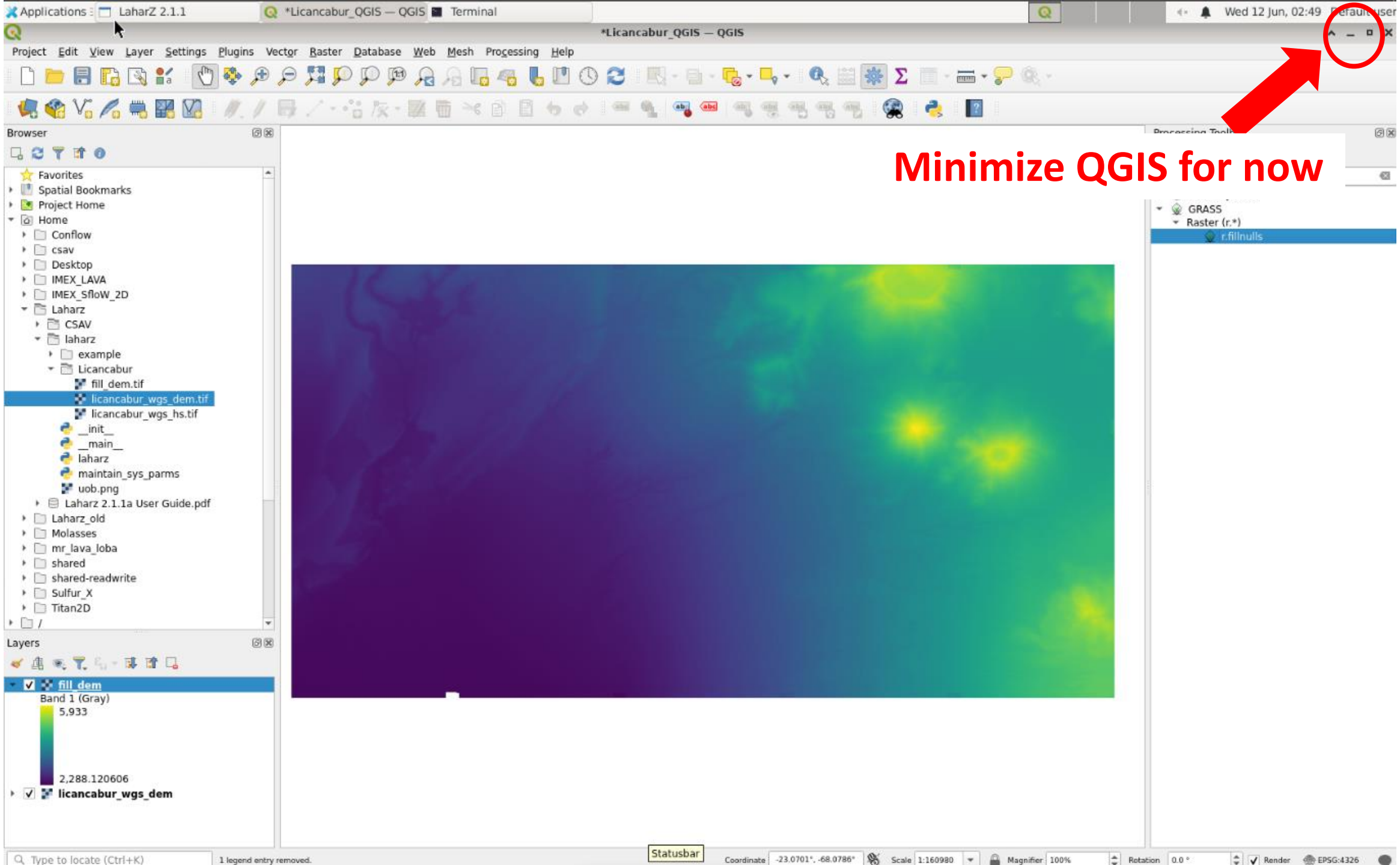
- **Navigate to the Laharz/Licancabur folder**
- **Drag and drop the "licancabur_wgs_dem.asc" DEM file onto your map to check that it looks good**

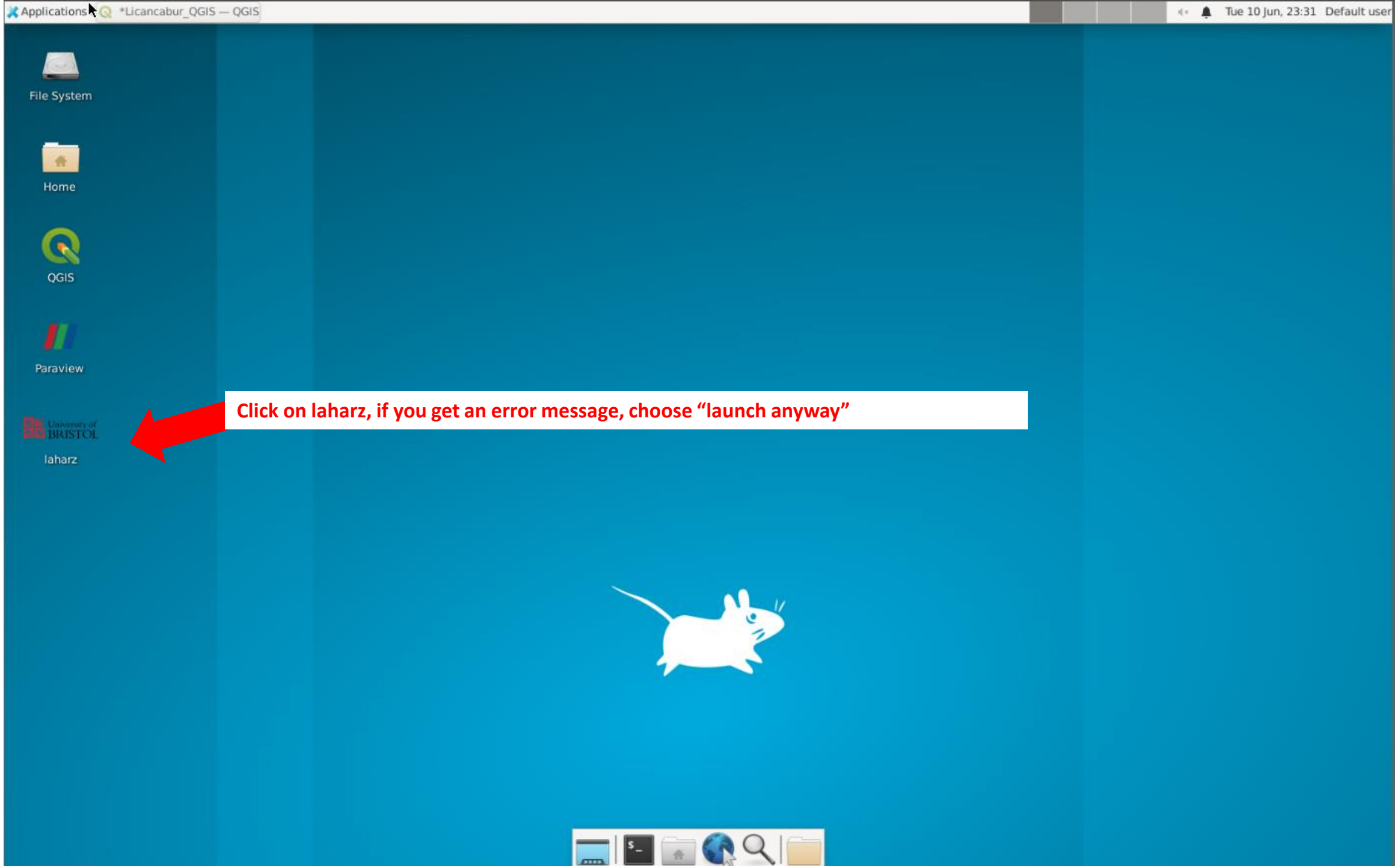


- Navigate to the DEM in the QGIS browser
- Right click on the DEM and choose “Export Layer”
- We will export the DEM as a GeoTiff. Make sure the fields look like the screenshot.
- Save the file in /LaharZ/Licancabur as “licancabur_wgs_dem.tif”



- **Now make a hillshade of the DEM**
- **Open the Raster Terrain Analysis →Hillshade tool**
- **Remember to change the Z factor to 0.00001 because the raster is in degrees lat/long**
- **Save the file in /LaharZ/Licancabur as “licancabur_wgs_hs.tif”**


Applications
LaharZ 2.1.1
*Licancabur_QGIS — QGIS
Terminal
Wed 12 Jun, 02:49
*Licancabur_QGIS — QGIS
Project Edit View Layer Settings Plugins Vector Raster Database Web Mesh Processing Help
Minimize QGIS for now
Browser
Favorites
Spatial Bookmarks
Project Home
Home
Conflow
csav
Desktop
IMEX_LAVA
IMEX_SfloW_2D
Laharz
CSAV
laharz
example
Licancabur
fill_dem.tif
licancabur_wgs_dem.tif
licancabur_wgs_hs.tif
__init__
__main__
laharz
maintain_sys_parms
uob.png
Laharz 2.1.1a User Guide.pdf
Laharz_old
Molasses
mr_lava_loba
shared
shared-readwrite
Sulfur_X
Titan2D
/
Layers
fill_dem
Band 1 (Gray)
5,933
2,288.120606
licancabur_wgs_dem
GRASS
Raster (r.*)
r.fillnulls
Type to locate (Ctrl+K)
1 legend entry removed.
Statusbar
Coordinate -23.0701°, -68.0786°
Scale 1:160980
Magnifier 100%
Rotation 0.0 °
Render
EPSG:4326


Applications
*Licancabur_QGIS — QGIS
Tue 10 Jun, 23:31
Default user
File System
Home
QGIS
Paraview
laharz
Click on laharz, if you get an error message, choose “launch anyway”

LaharZ 2.1.4

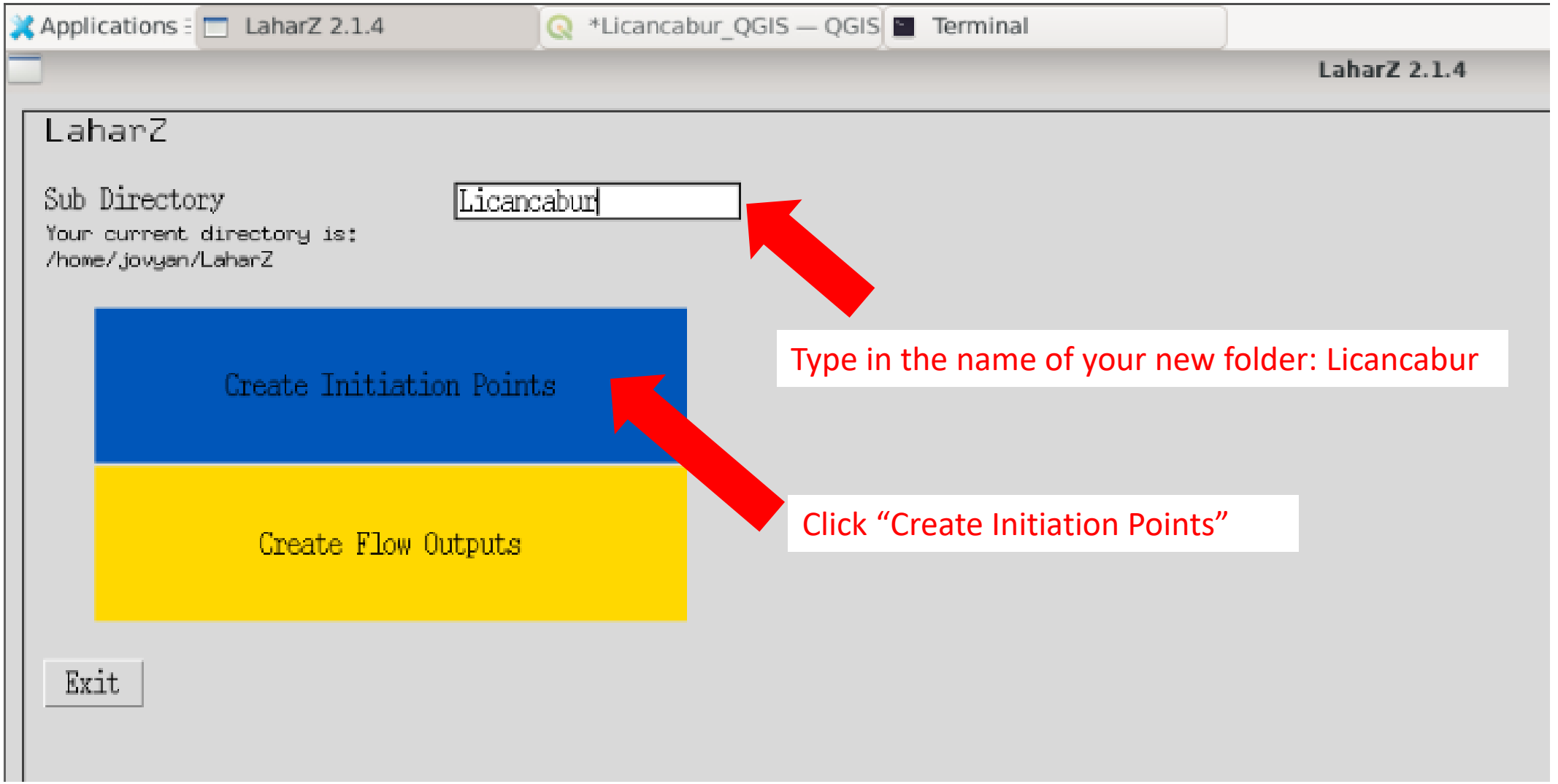
LaharZ



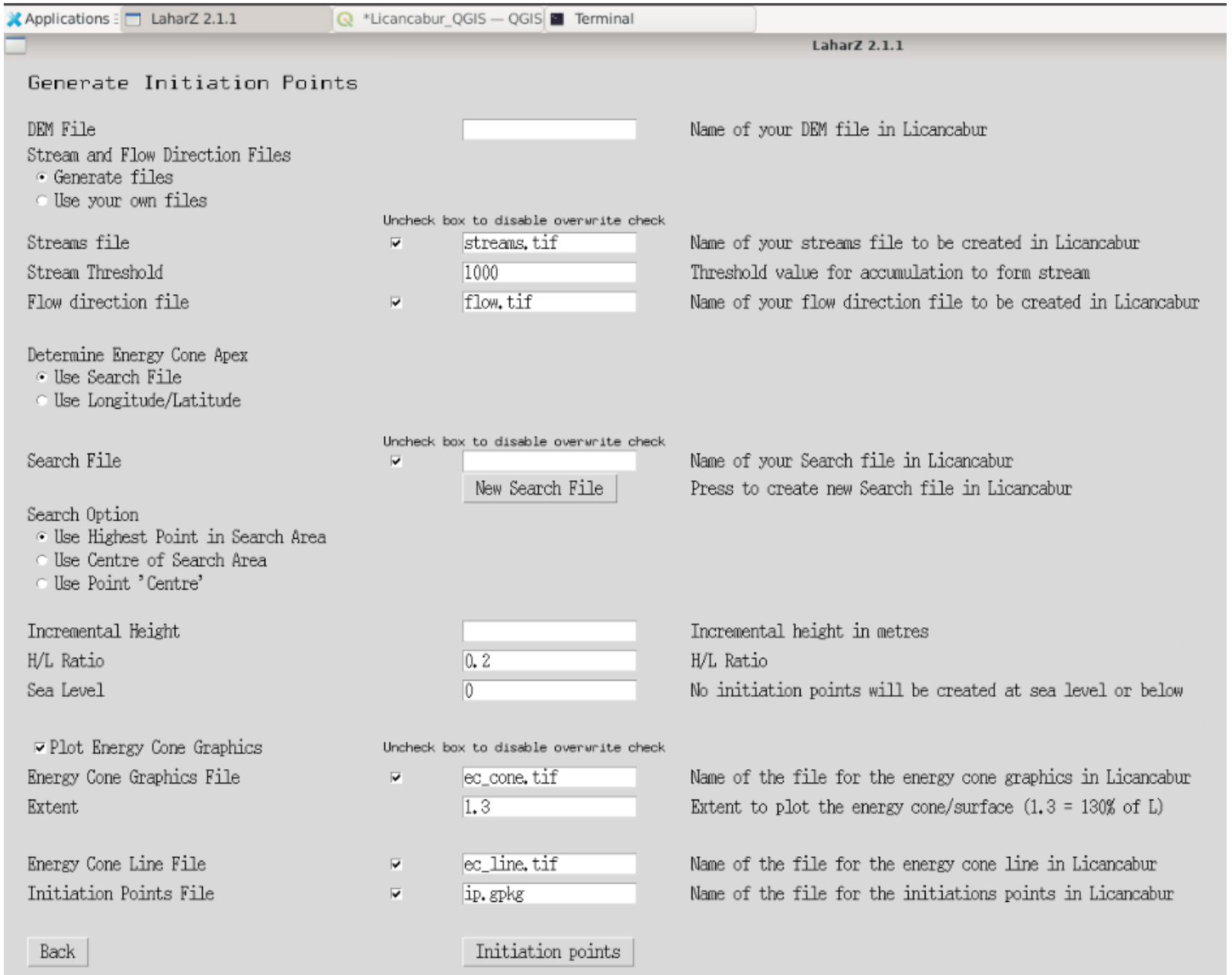
University of BRISTOL

Version: 2.1.4 stand alone

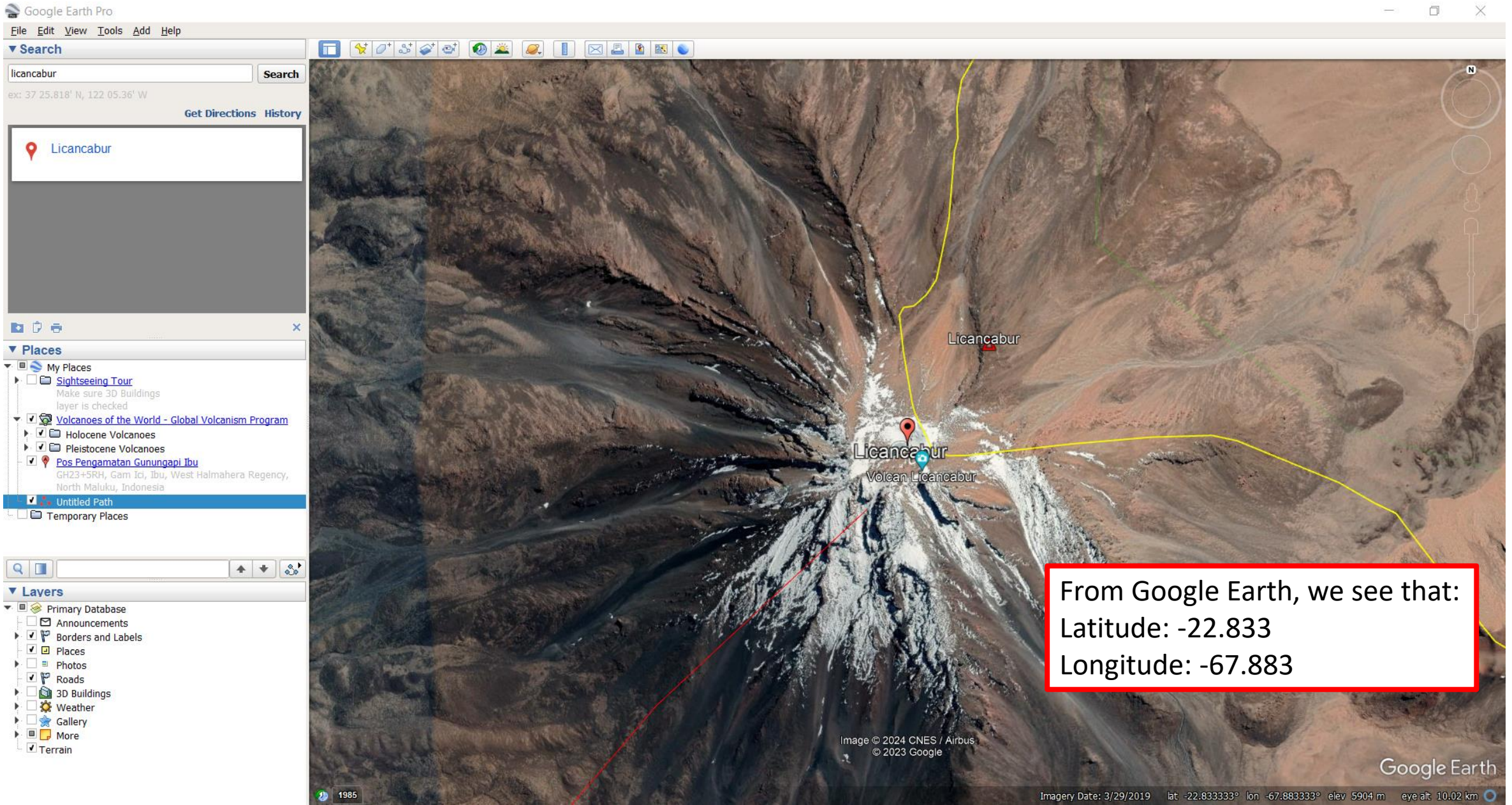
Let’s make a stream network, an H/L cone, and start points

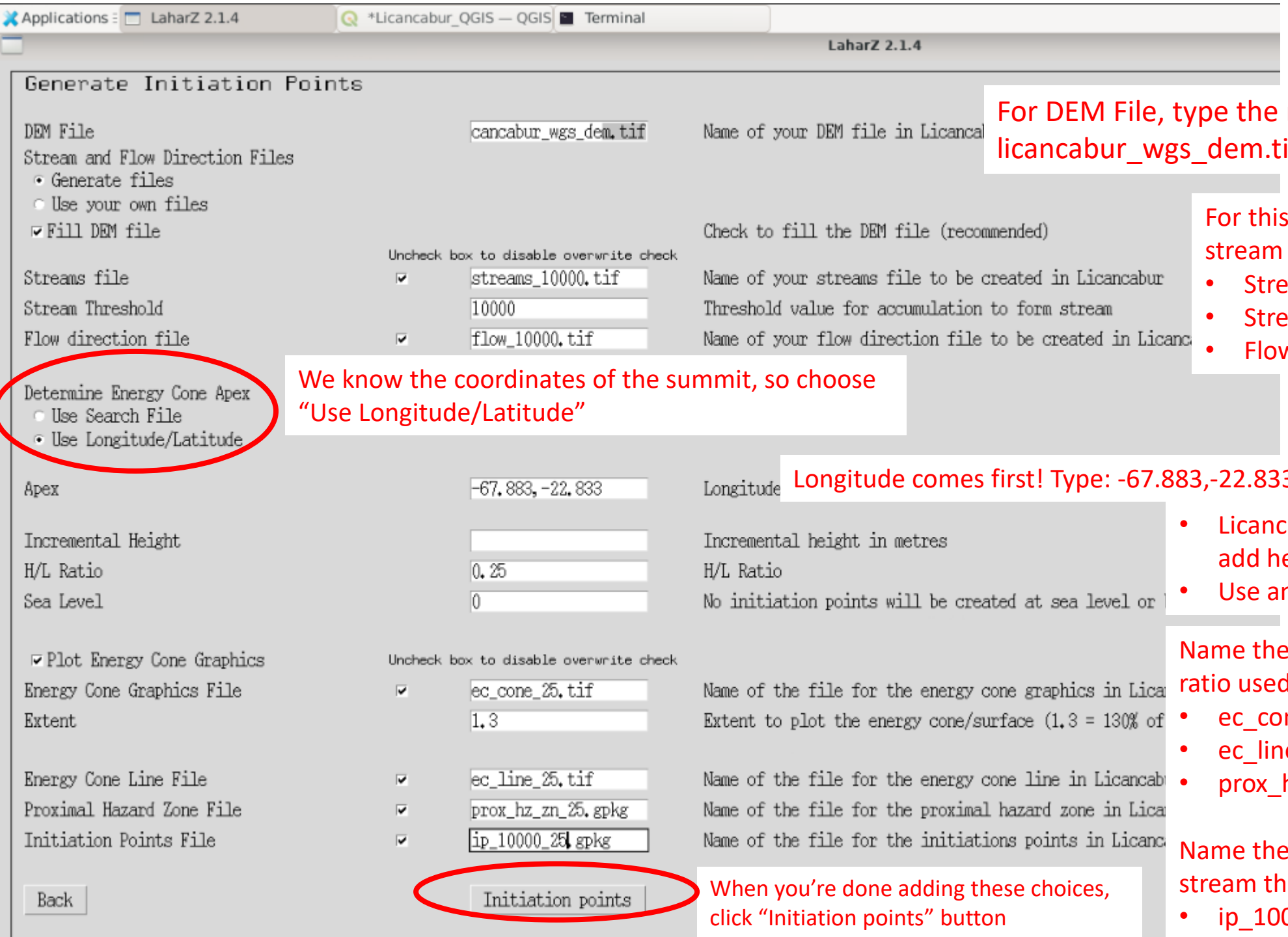


We have some fields to fill in! There are some default values to change.



For this exercise, we need coordinates of the summit





For DEM File, type the name of your filled DEM: licancabur_wgs_dem.tif

For this step, we're going to try a stream threshold of 10000.

- Streams file: streams_10000.tif
- Stream threshold: 10000
- Flow direction file: flow_10000.tif

We know the coordinates of the summit, so choose "Use Longitude/Latitude"

Longitude comes first! Type: -67.883,-22.833

- Licancabur is nice and conical, so don't add height
- Use an H/L ratio of 0.25

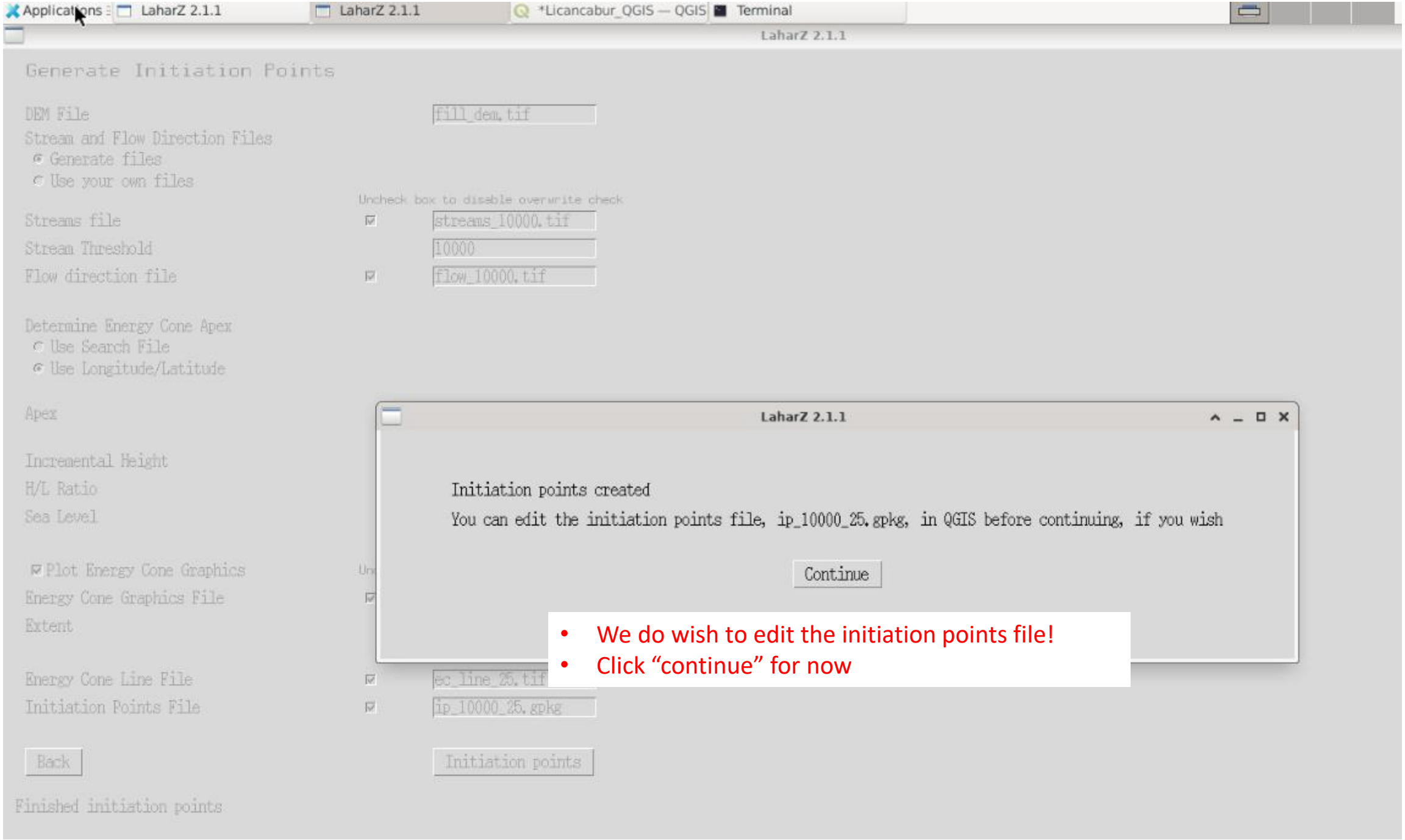
Name the energy cone files with the H/L ratio used:

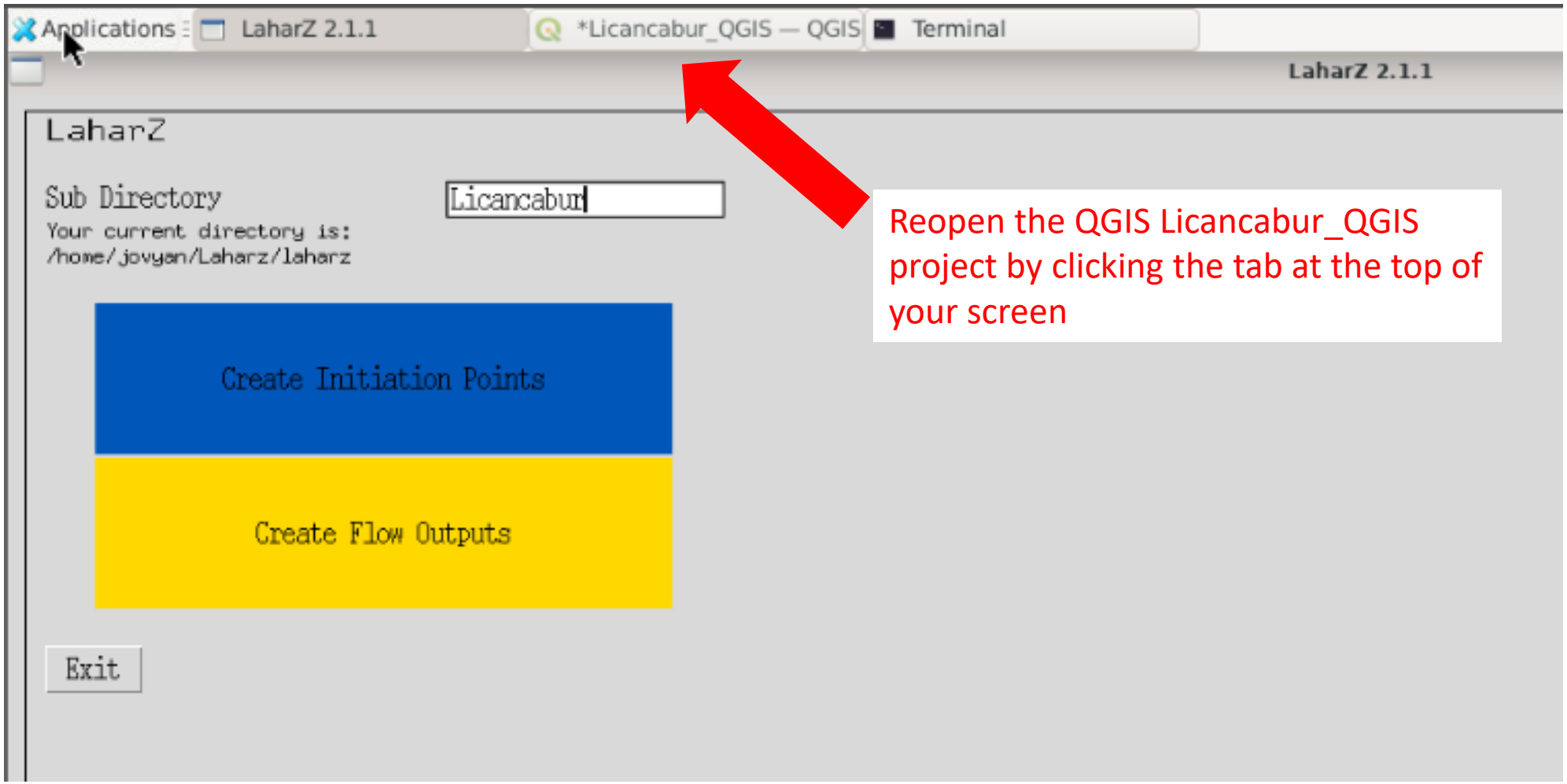
- ec_cone_25.tif
- ec_line_25.tif
- prox_hz_zn_25.gpkg

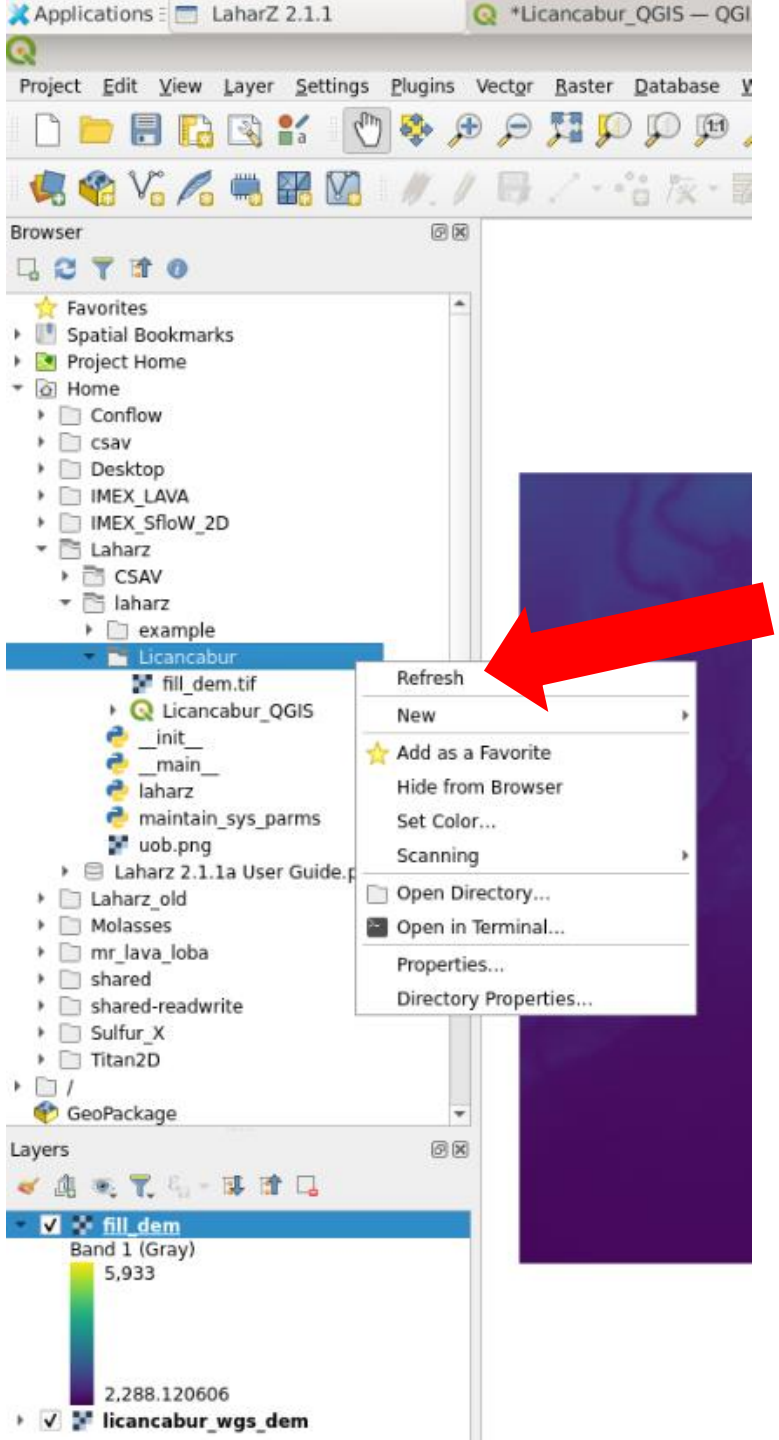
Name the initiation points with the stream threshold and H/L ratio:

- ip_10000_25.gpkg

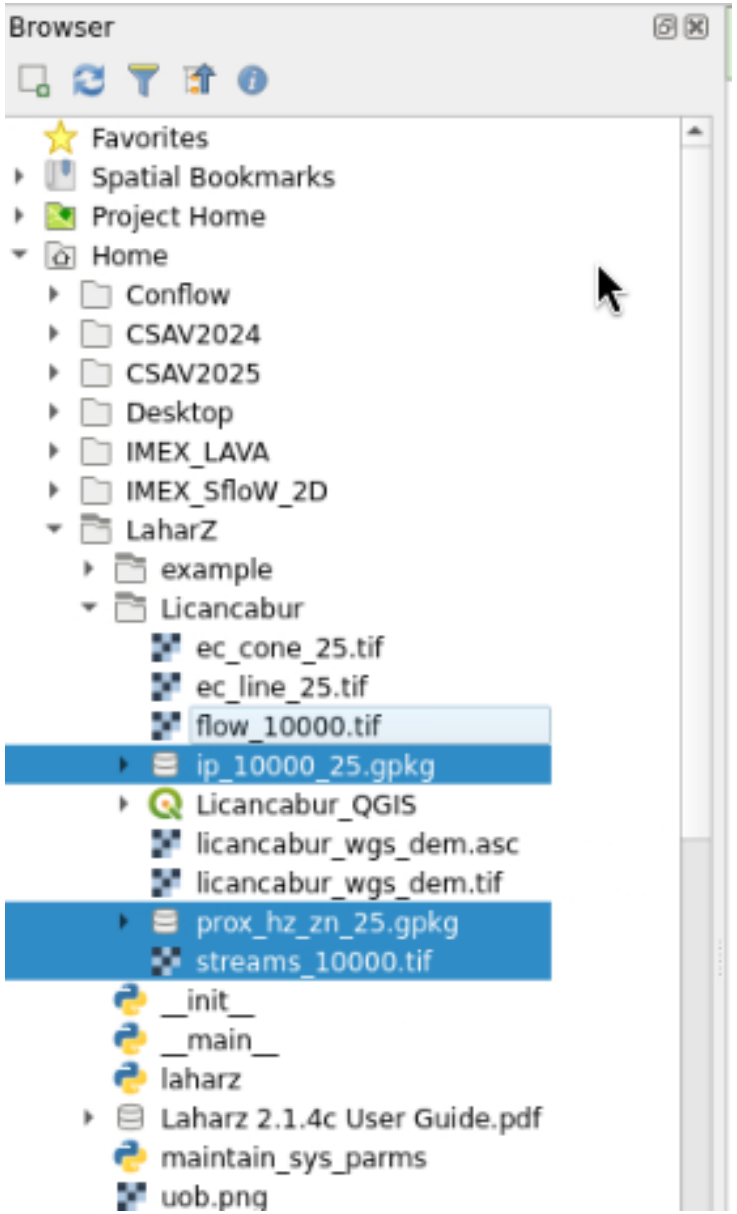
When you're done adding these choices, click "Initiation points" button


Applications
LaharZ 2.1.1
LaharZ 2.1.1
*Licancabur_QGIS — QGIS
Terminal
LaharZ 2.1.1
Generate Initiation Points
DEM File
fill_dem.tif
Stream and Flow Direction Files
Generate files
Use your own files
Uncheck box to disable overwrite check
Streams file
streams_10000.tif
Stream Threshold
10000
Flow direction file
flow_10000.tif
Determine Energy Cone Apex
Use Search File
Use Longitude/Latitude
Apex
Incremental Height
H/L Ratio
Sea Level
Plot Energy Cone Graphics
Energy Cone Graphics File
Extent
Energy Cone Line File
ec_line_25.tif
Initiation Points File
ip_10000_25.gpkg
Back
Initiation points
Finished initiation points
LaharZ 2.1.1
Initiation points created
You can edit the initiation points file, ip_10000_25.gpkg, in QGIS before continuing, if you wish
Continue
• We do wish to edit the initiation points file!
• Click “continue” for now


Applications
LaharZ 2.1.1
*Licancabur_QGIS — QGIS
Terminal
LaharZ 2.1.1
LaharZ
Sub Directory
Licancabur
Your current directory is:
/home/jovyan/Laharz/laharz
Create Initiation Points
Create Flow Outputs
Exit
Reopen the QGIS Licancabur_QGIS project by clicking the tab at the top of your screen

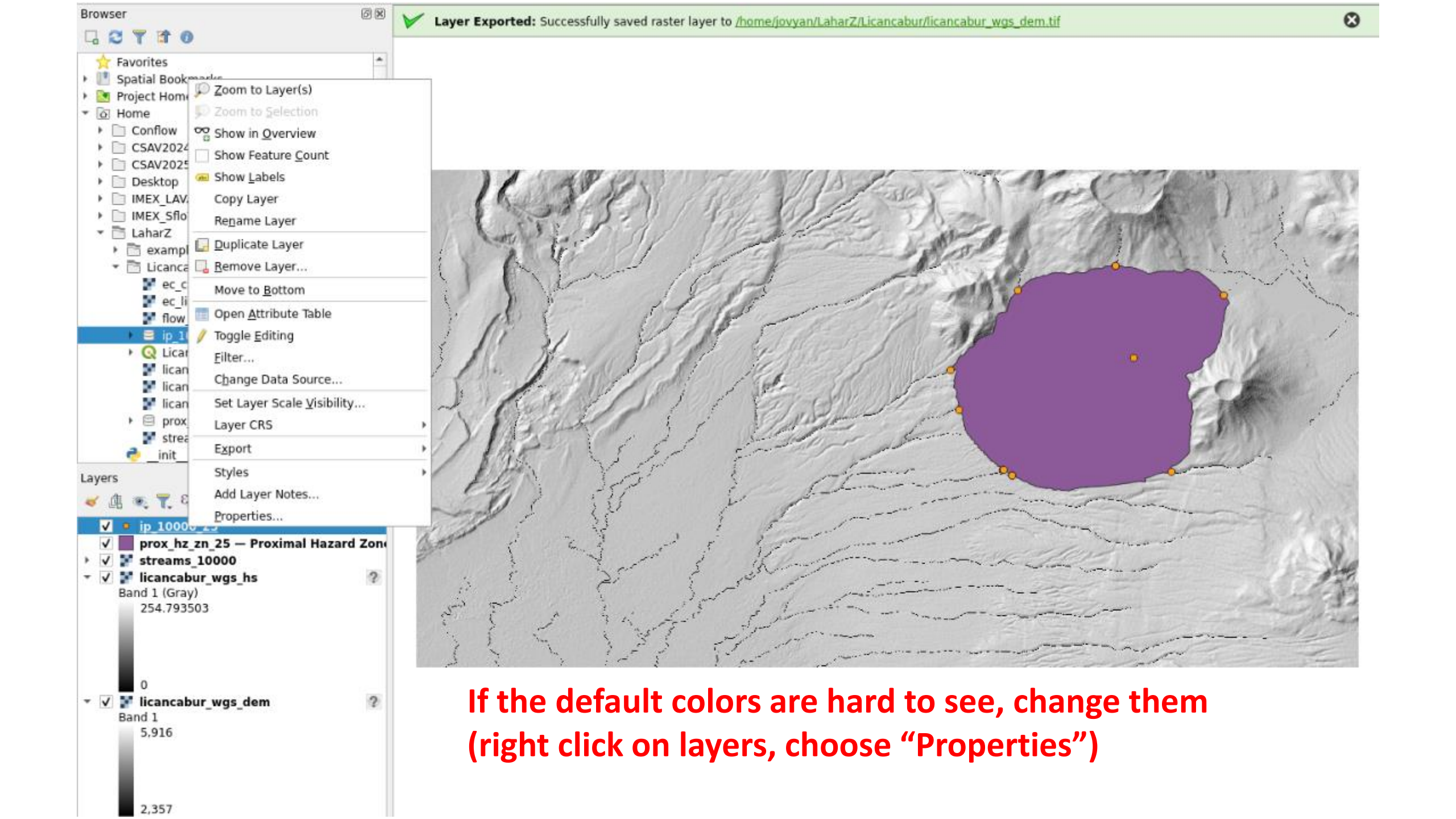


Right click on the “Licancabur” folder and click “Refresh” to see your new files

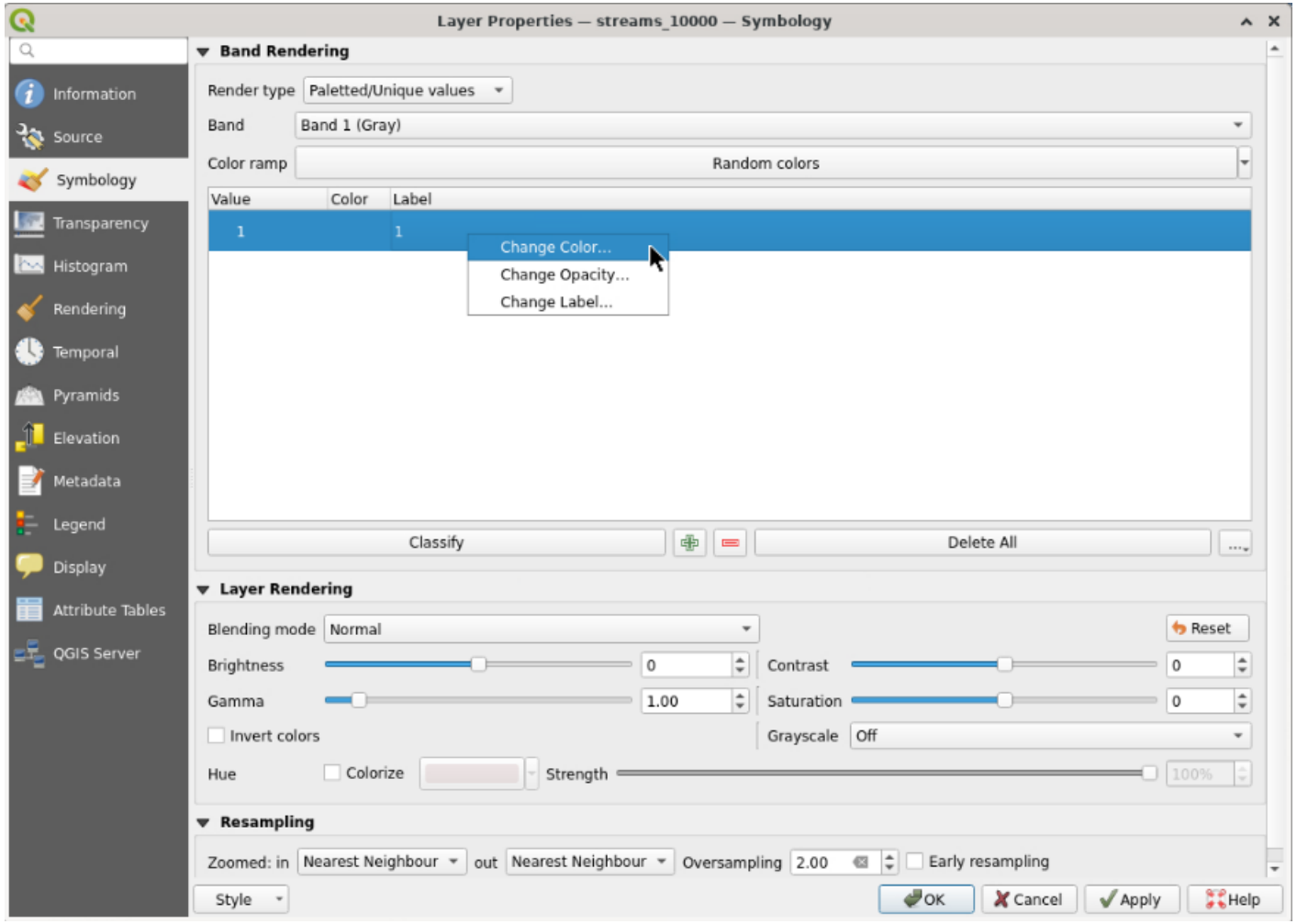


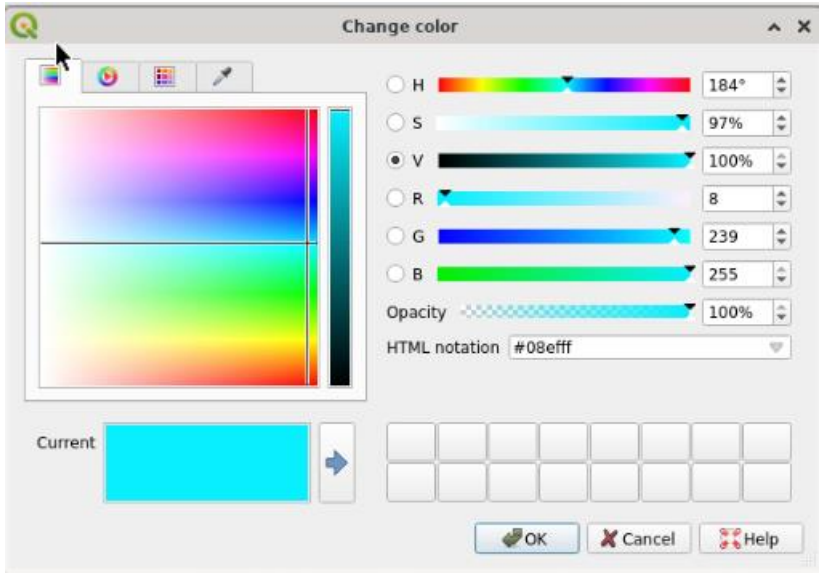
Let's look at the stream network, H/L cone (ec_line), and the initiation points we created!

Add these files to your map:
streams_10000.tif
ec_line_25.tif
ip_10000_25.gpkg
prox_hz_zn_25.gpkg

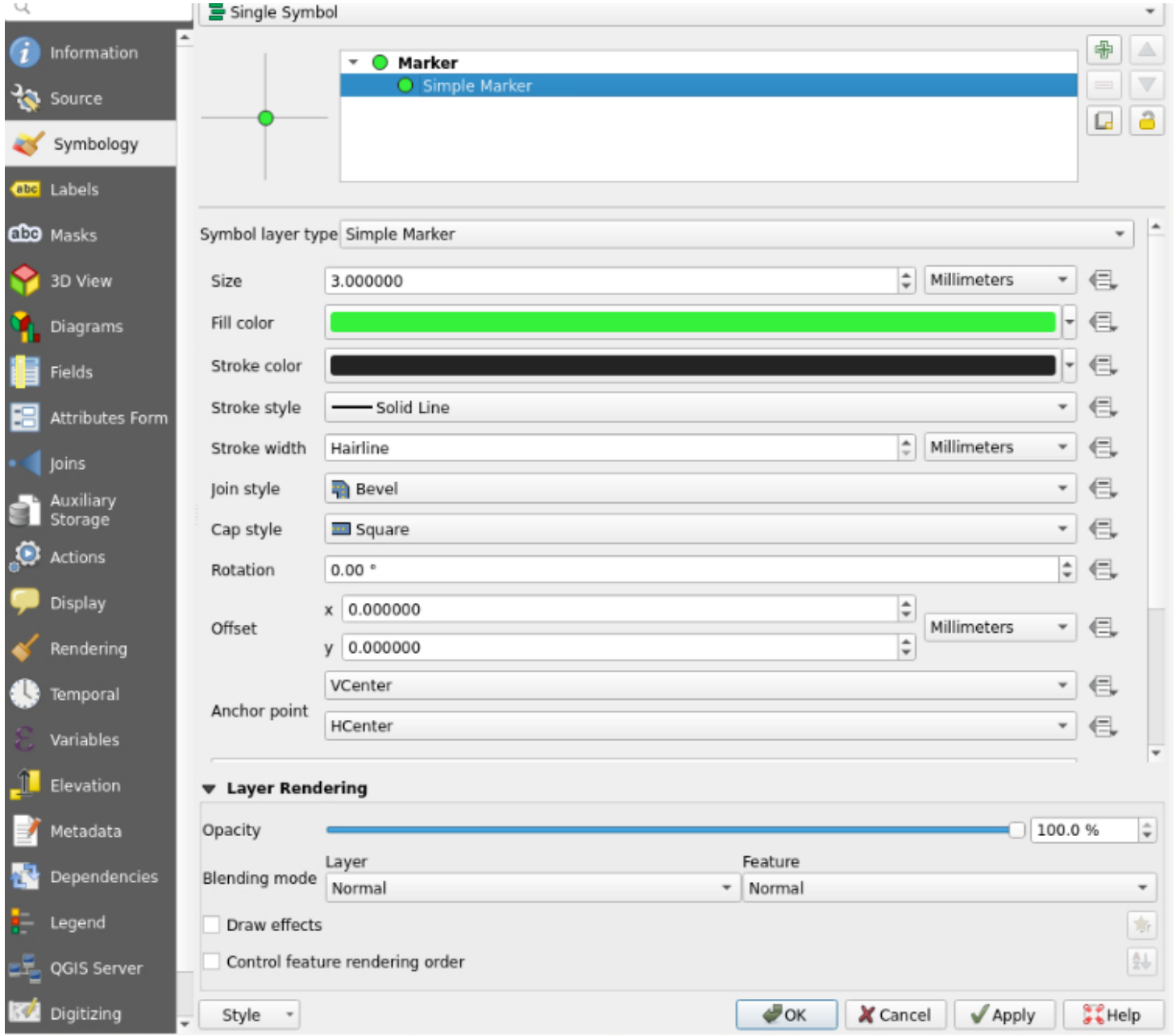


If the default colors are hard to see, change them (right click on layers, choose “Properties”)



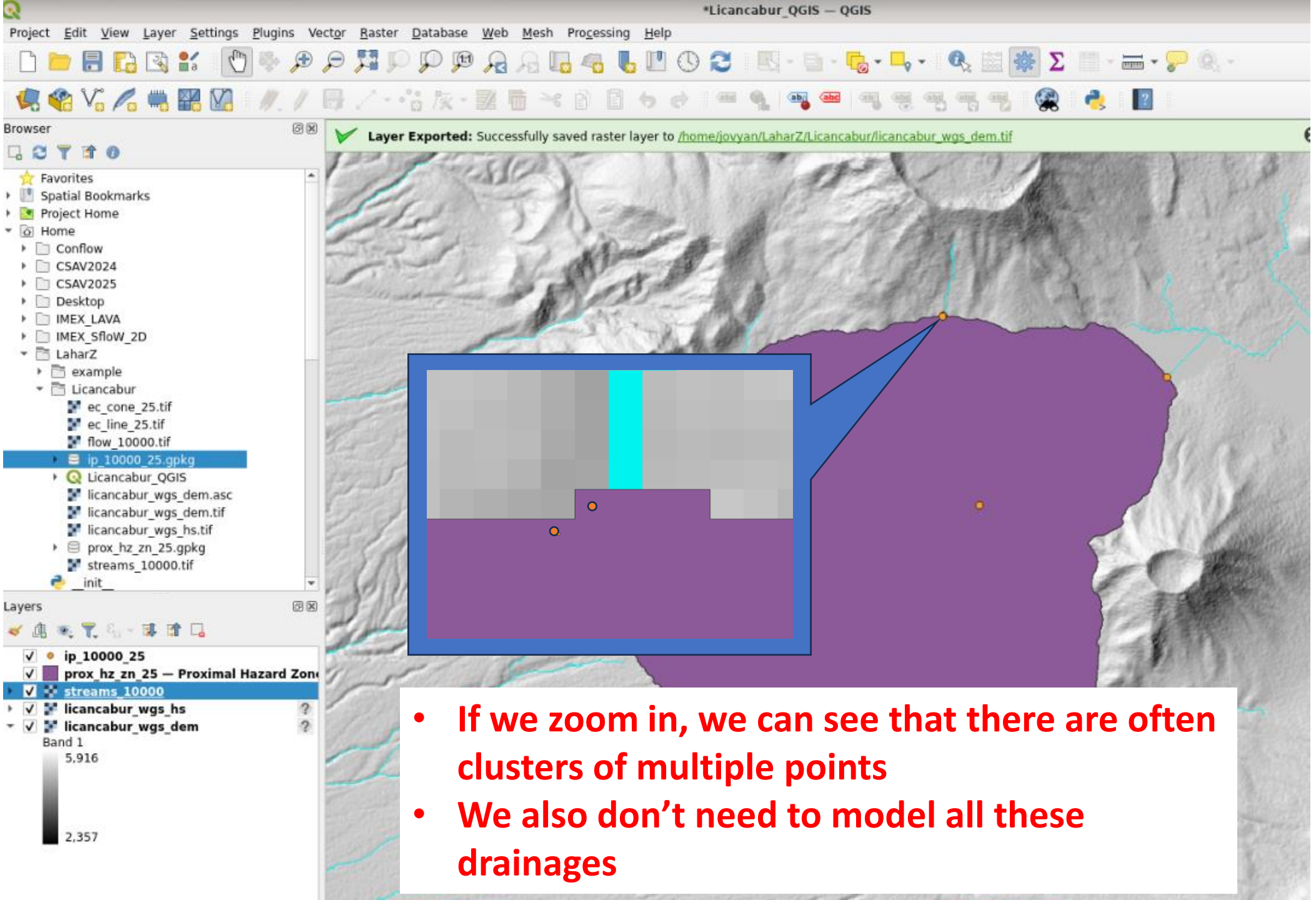


For the stream network change the “render type” to “Paletted/unique values” and click “classify”. Then right click on the listed value and choose “change color”

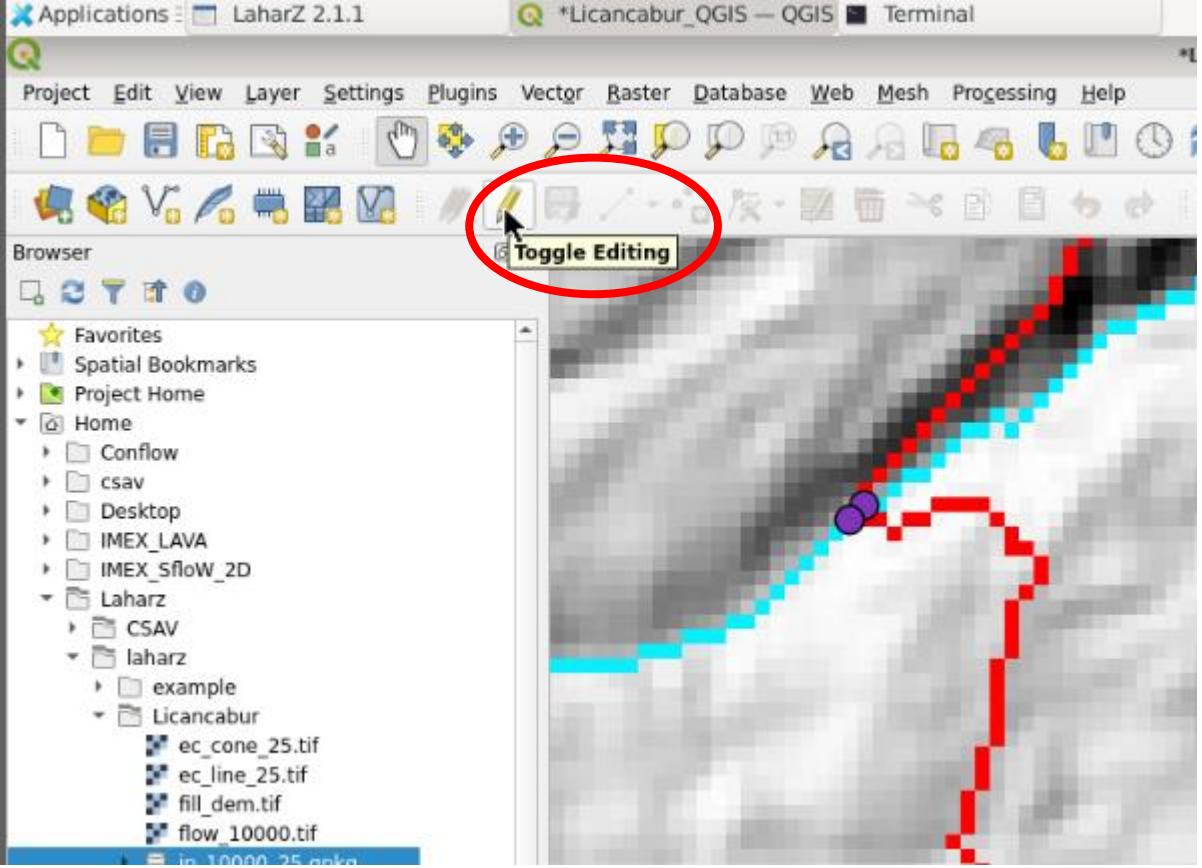


For the initiation points, change the color to something you like.

You can also change the size of the points if you want.

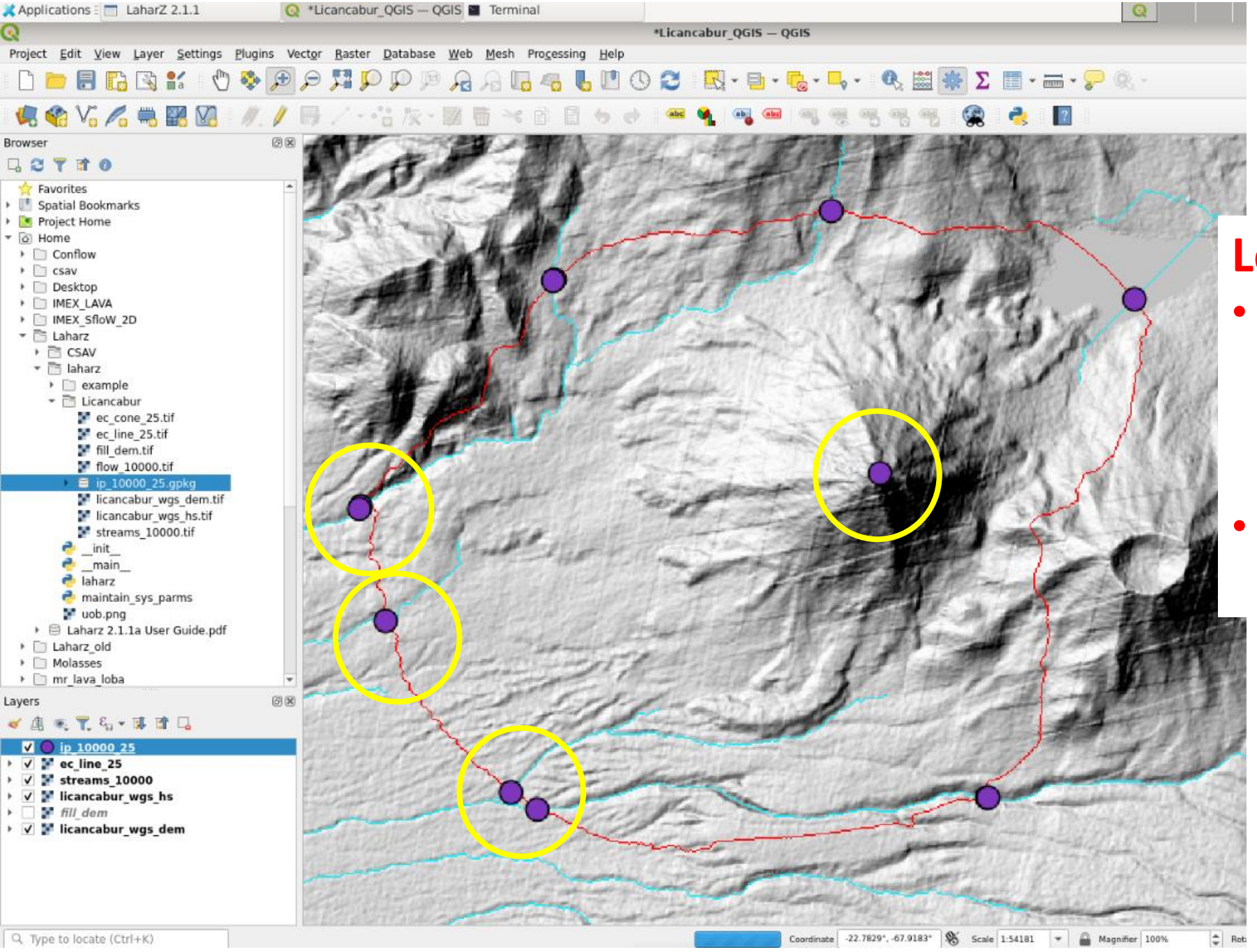


- If we zoom in, we can see that there are often clusters of multiple points
- We also don't need to model all these drainages



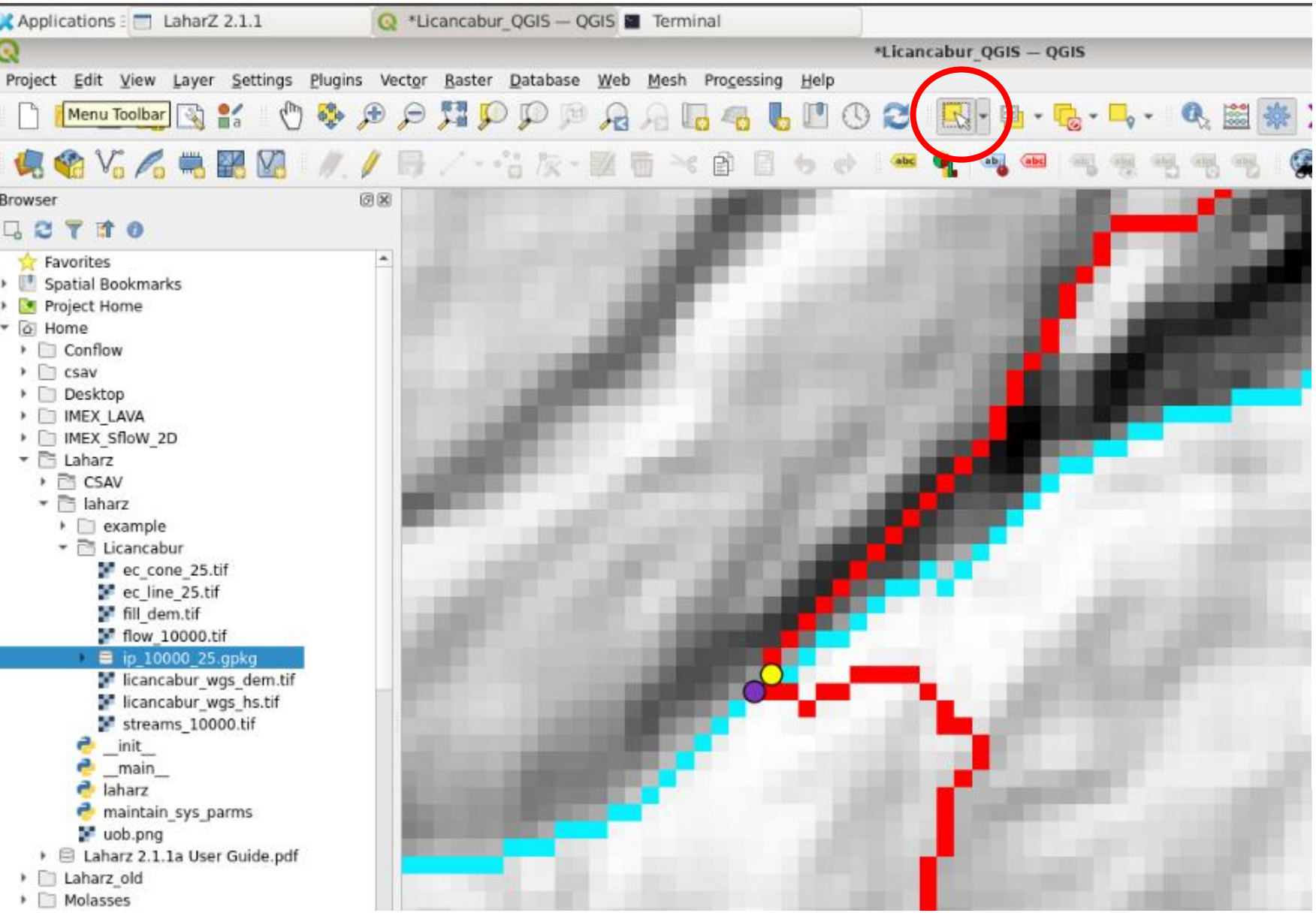
Let's delete some points

- Turn editing on by clicking the pencil "Toggle editing"



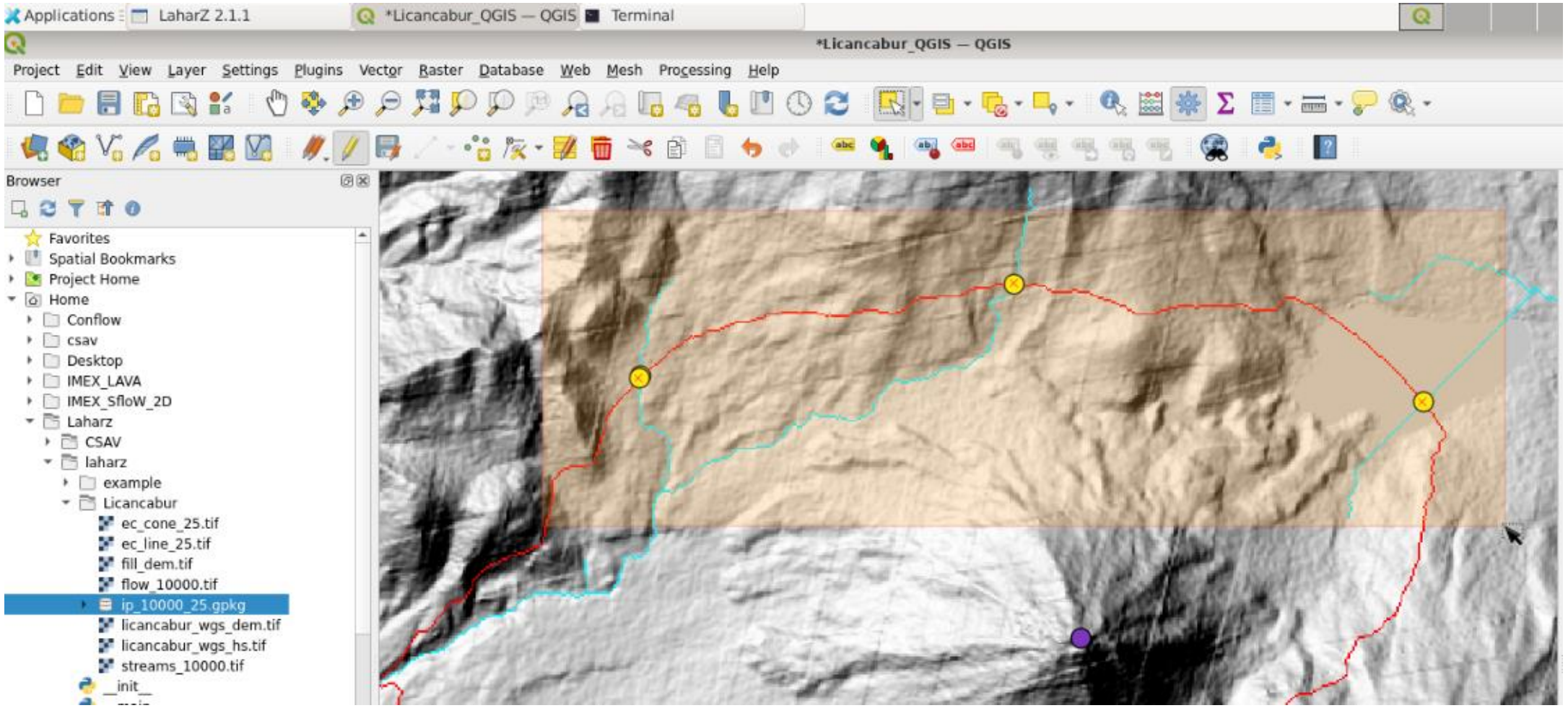
Let's delete some points

- Let's keep just one point in each of the areas circled and delete everything else
- Also keep the summit point



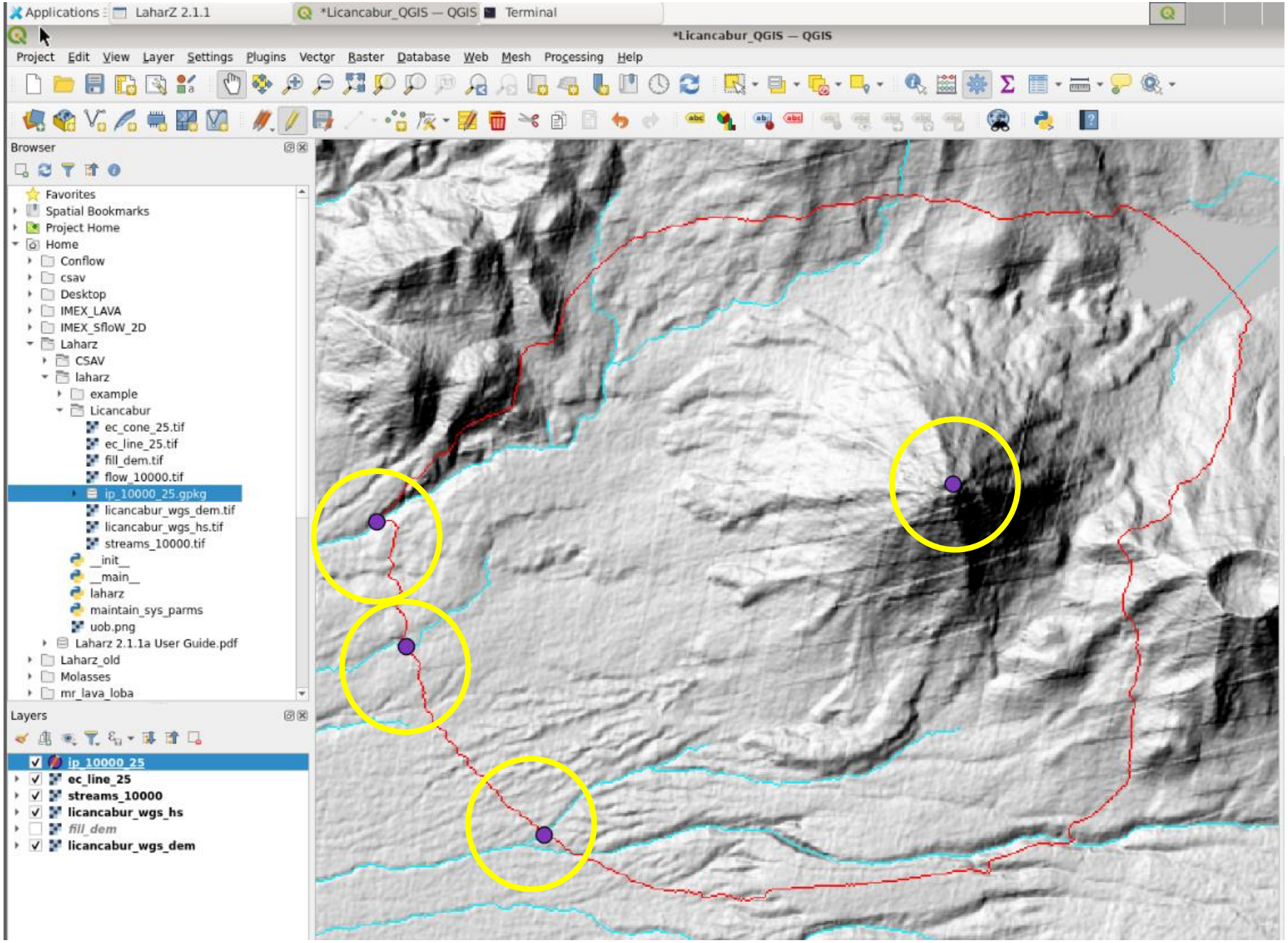
Let's delete some points

- Turn on the "Select features by area" tool and zoom in to each point cluster
- Click on the point you want to delete
- It will turn yellow
- The press the "delete" key on your keyboard

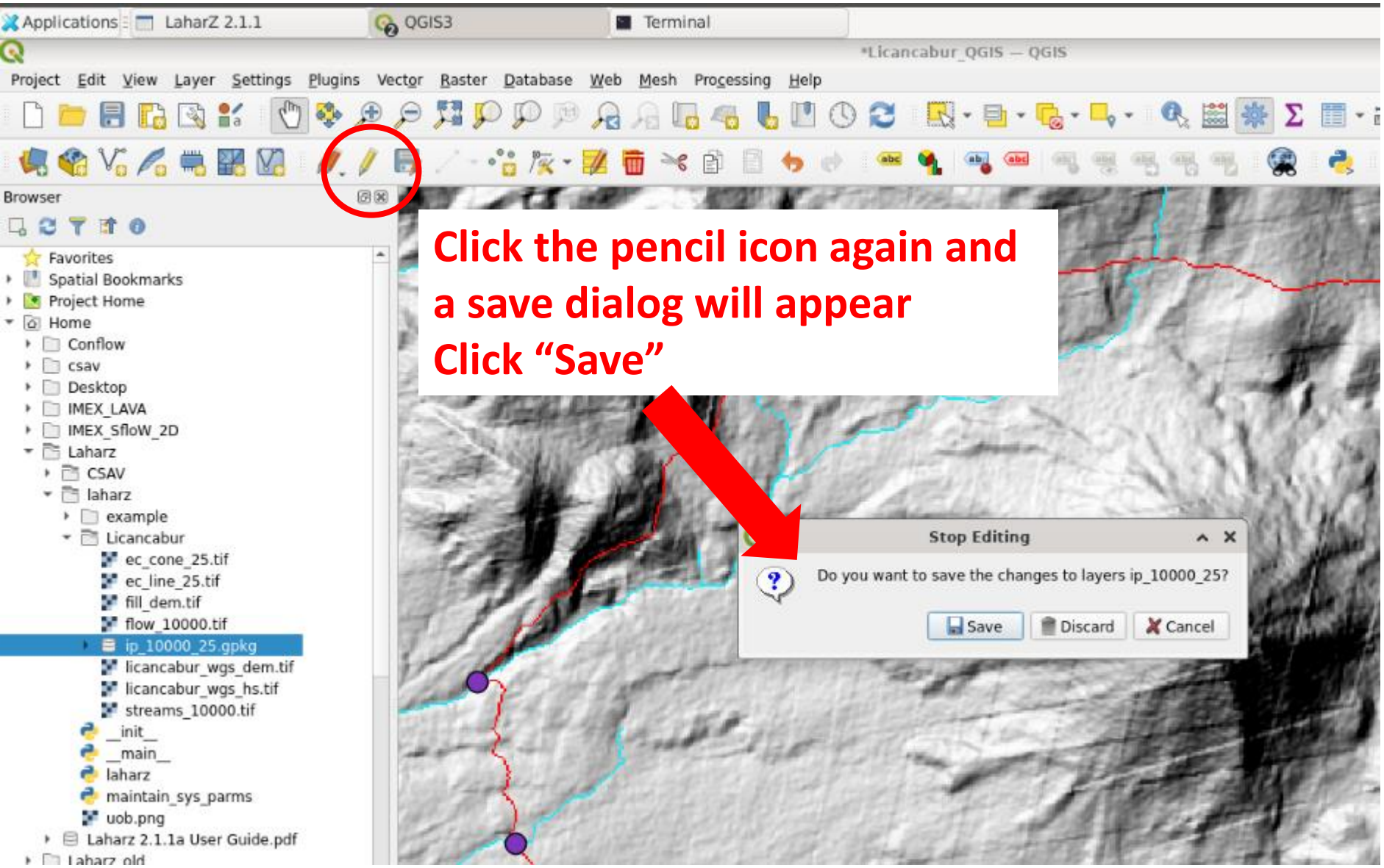


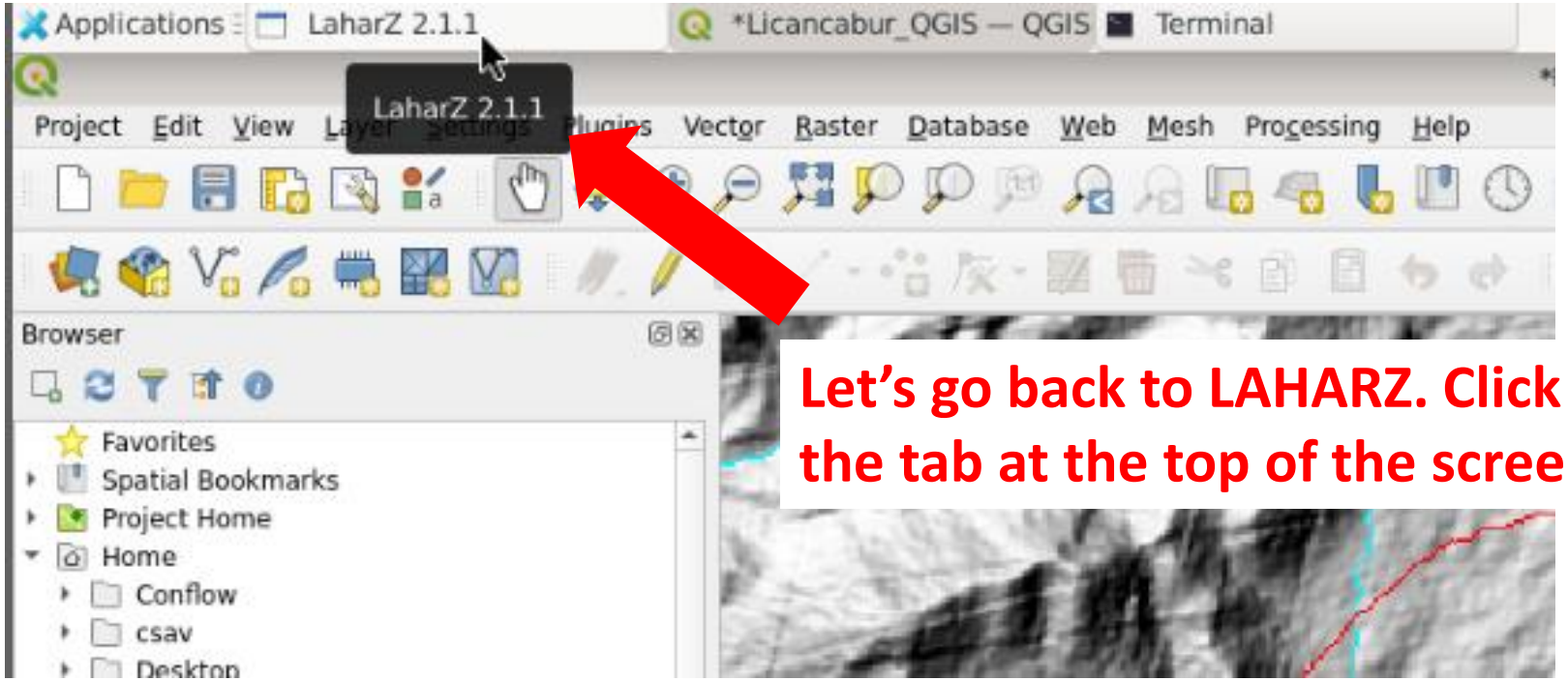
Let's delete some points

- You can also use the "Select features by area" tool to drag a selection box around multiple points
- They will turn yellow
- The press the "delete" key on your keyboard

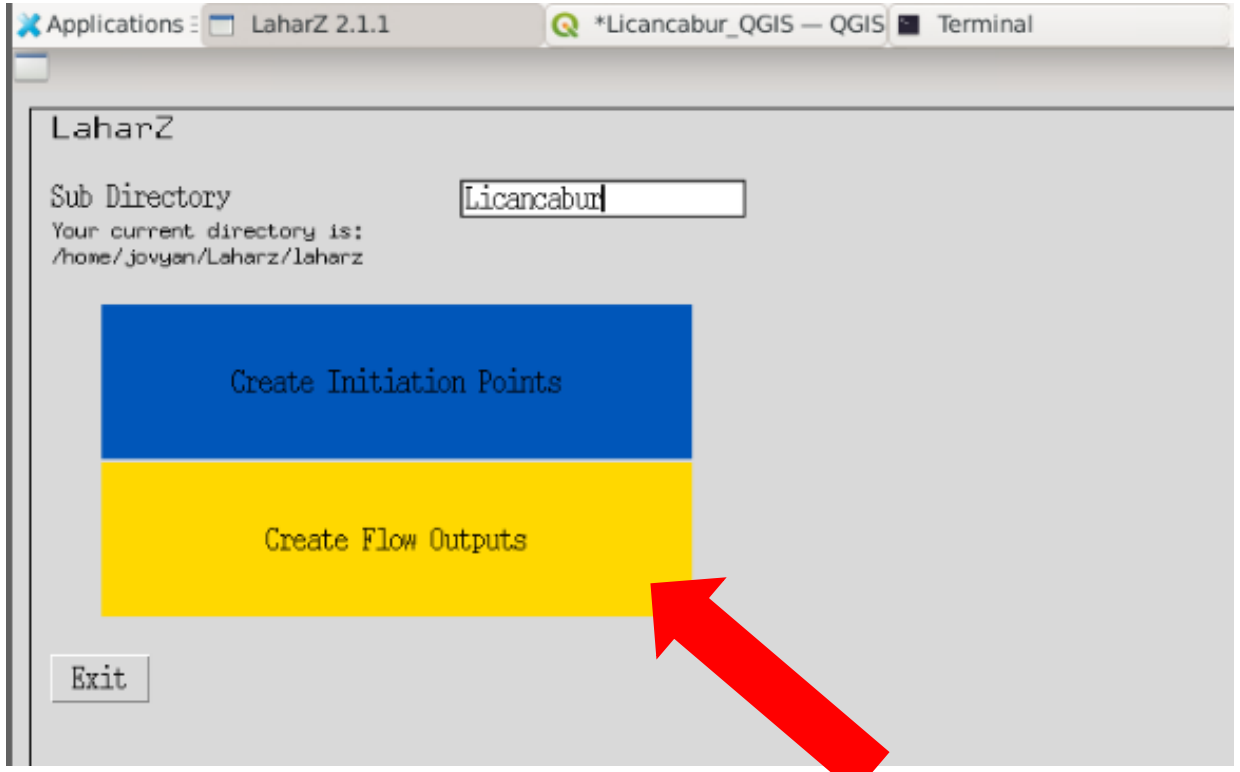


- We should now have 3 start points and 1 summit point left
- Click the pencil icon again and a save dialog will appear

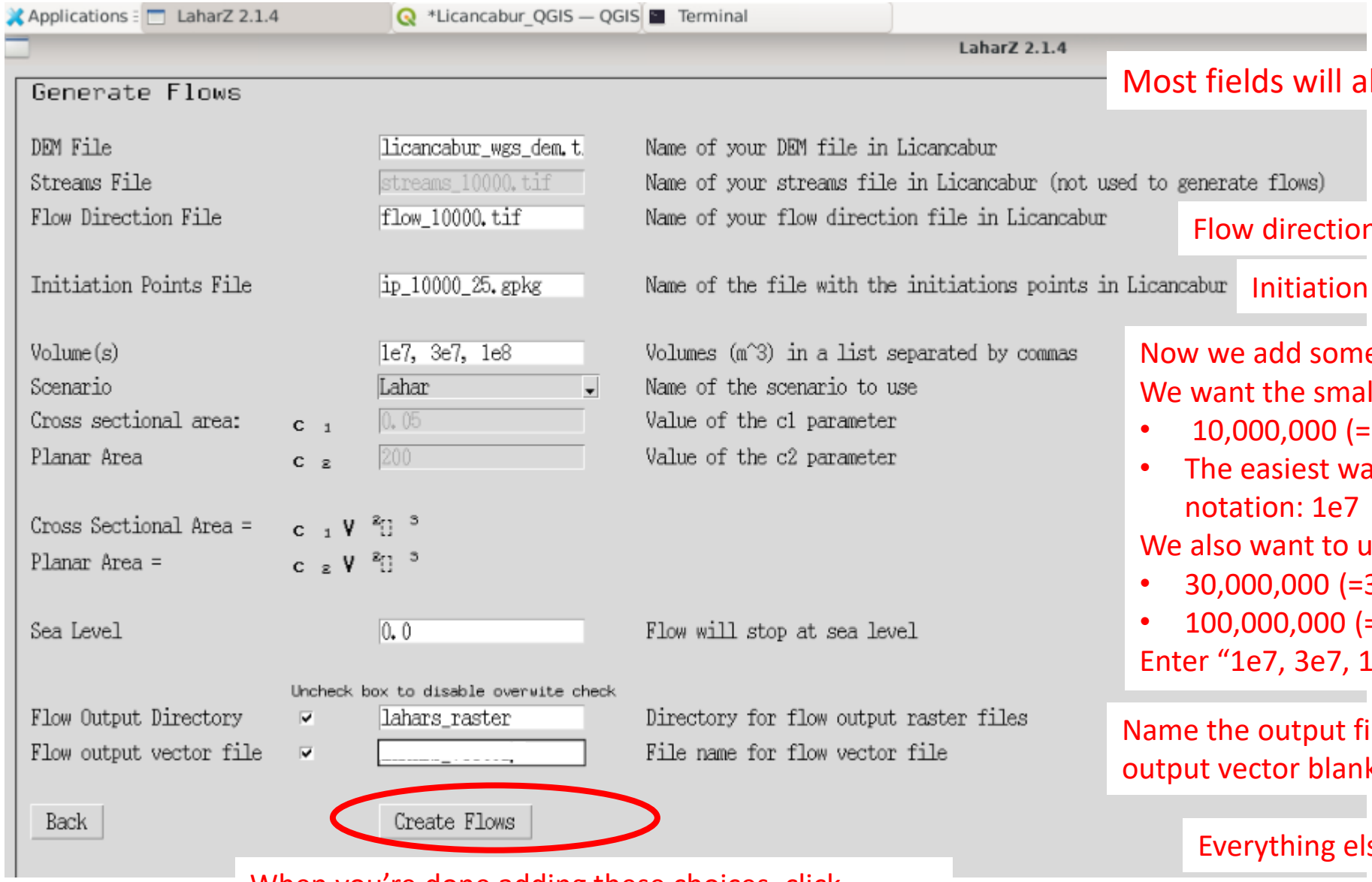

Applications
LaharZ 2.1.1
QGIS3
Terminal
*Licancabur_QGIS — QGIS
Project Edit View Layer Settings Plugins Vector Raster Database Web Mesh Processing Help
Browser
Favorites
Spatial Bookmarks
Project Home
Home
Conflow
csav
Desktop
IMEX_LAVA
IMEX_SfloW_2D
Laharz
CSAV
laharz
example
Licancabur
ec_cone_25.tif
ec_line_25.tif
fill_dem.tif
flow_10000.tif
ip_10000_25.gpkg
licancabur_wgs_dem.tif
licancabur_wgs_hs.tif
streams_10000.tif
__init__
__main__
laharz
maintain_sys_parms
uob.png
Laharz 2.1.1a User Guide.pdf
Laharz_old
Click the pencil icon again and
a save dialog will appear
Click “Save”
Stop Editing
Do you want to save the changes to layers ip_10000_25?
Save
Discard
Cancel


Applications
LaharZ 2.1.1
*Licancabur_QGIS — QGIS
Terminal
LaharZ 2.1.1
Project Edit View Layer Settings Plugins Vector Raster Database Web Mesh Processing Help
Browser
Favorites
Spatial Bookmarks
Project Home
Home
Conflow
csav
Desktop
Let’s go back to LAHARZ. Click the tab at the top of the screen

Let’s run LAHARZ!



Click “Create Flow Outputs”



Most fields will already be filled in

Flow direction file: flow_10000.tif

Initiation points file: ip_10000_25.gpkg

Now we add some volumes

We want the smallest volume to be:

- 10,000,000 (=10 million m^3=0.01km^3=1x10^7=1e7)
- The easiest way to enter volumes is using scientific notation: 1e7

We also want to use:

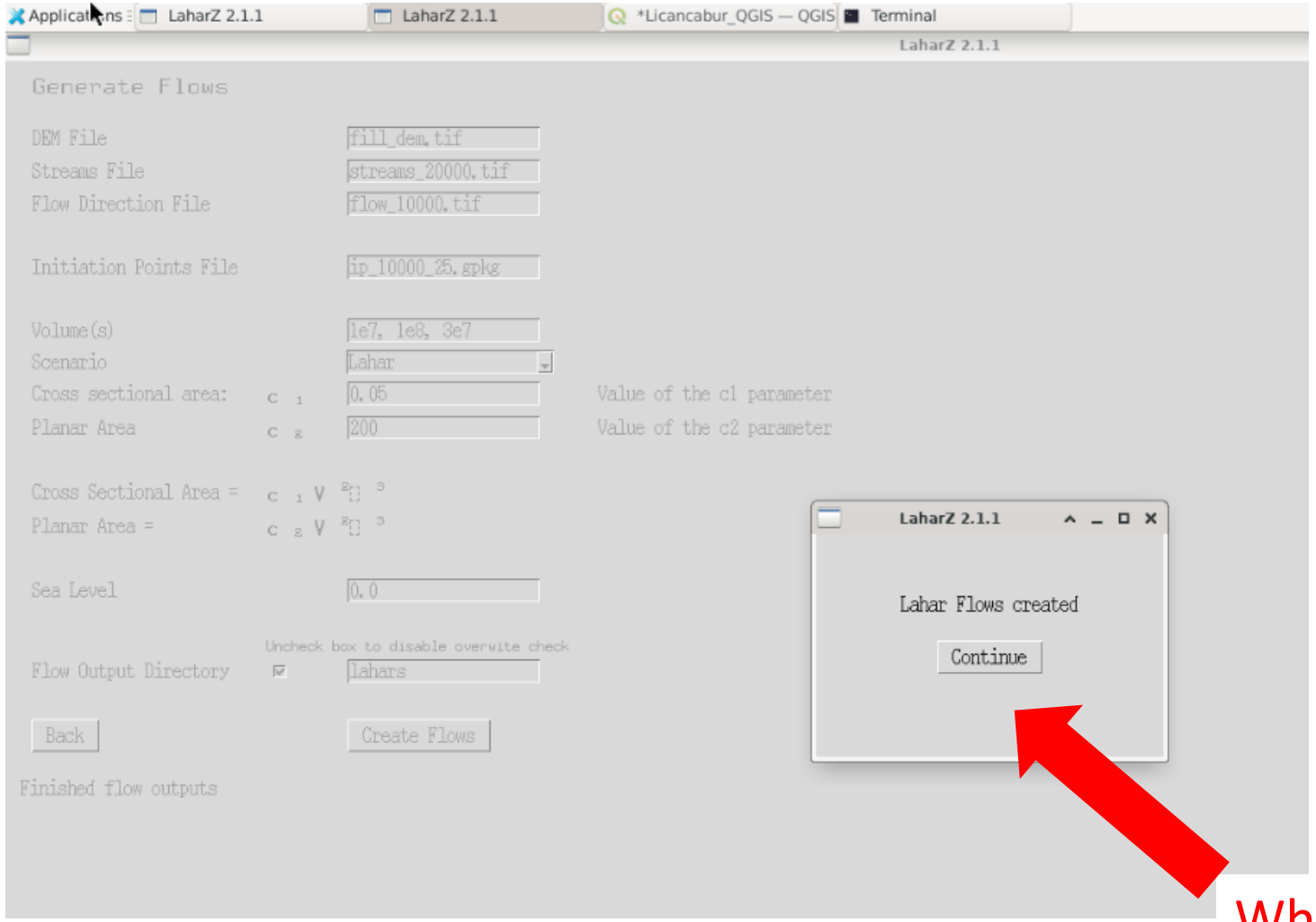
- 30,000,000 (=30 million m^3=0.03 km^3=3x10^7=3e7)
- 100,000,000 (=100 million m^3=0.1km^3=1x10^8=1e8)

Enter “1e7, 3e7, 1e8” in the Volumes box

Name the output files (for this exercise, we’ll leave output vector blank)

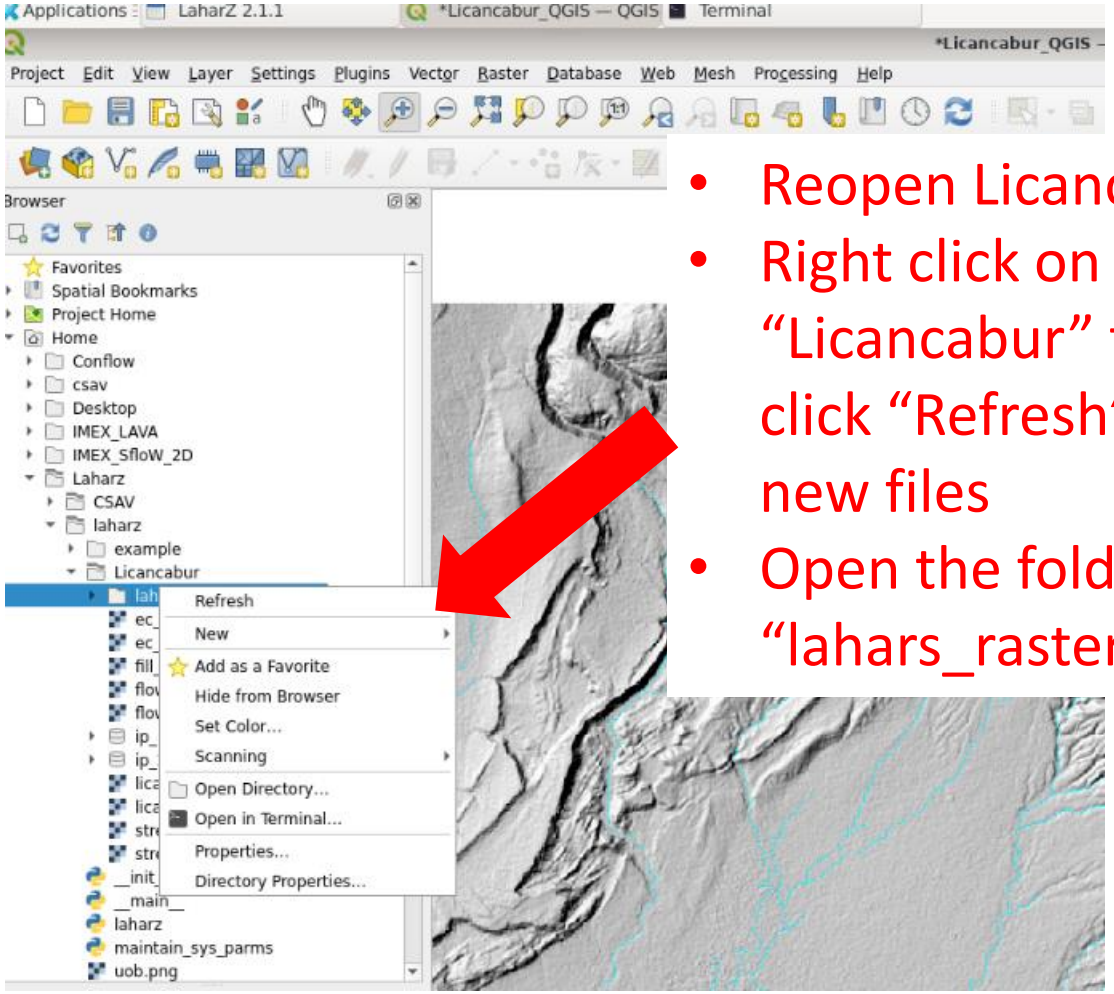
Everything else can stay as the default

When you’re done adding these choices, click “Create flows” button

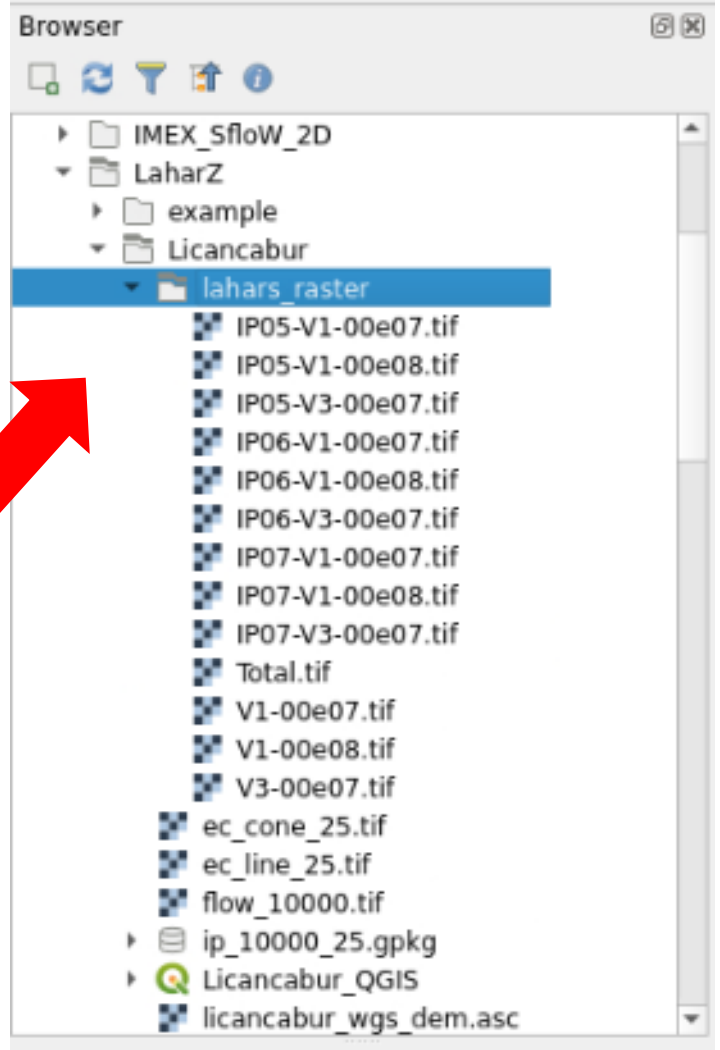


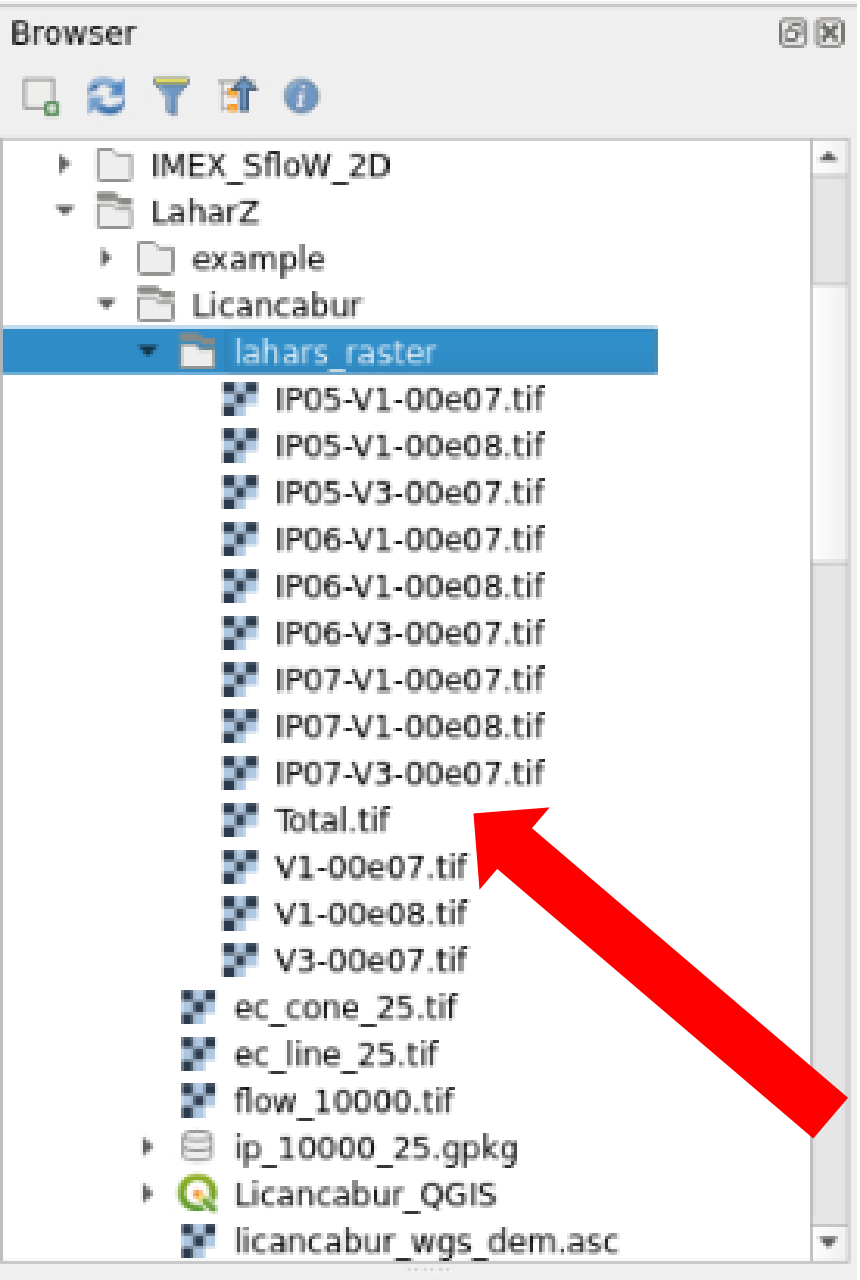
When it’s done, click “Continue”

Let’s look at the new files.



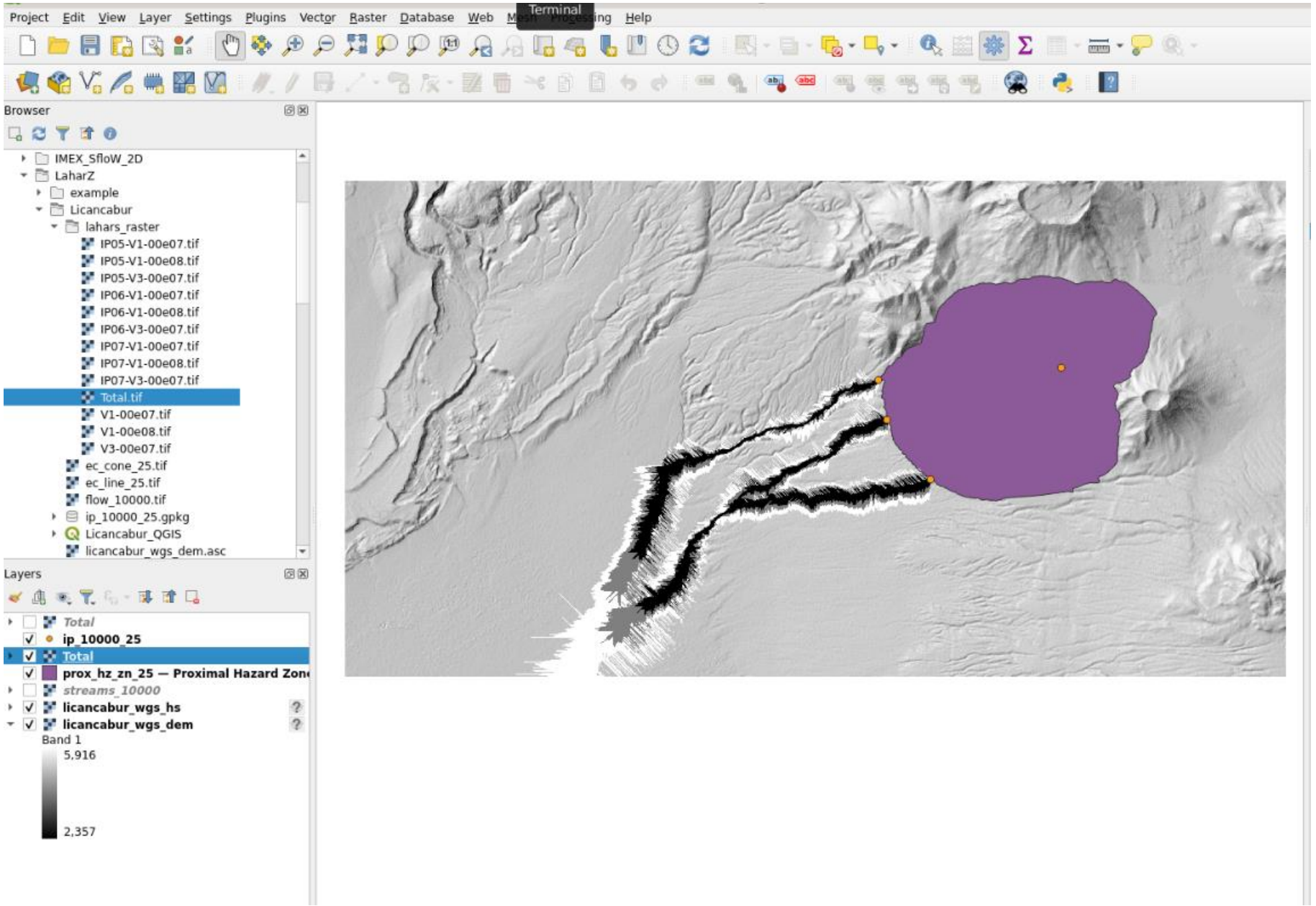
- Reopen Licancabur_QGIS
- Right click on the “Licancabur” folder and click “Refresh” to see your new files
- Open the folder called “lahars_raster”

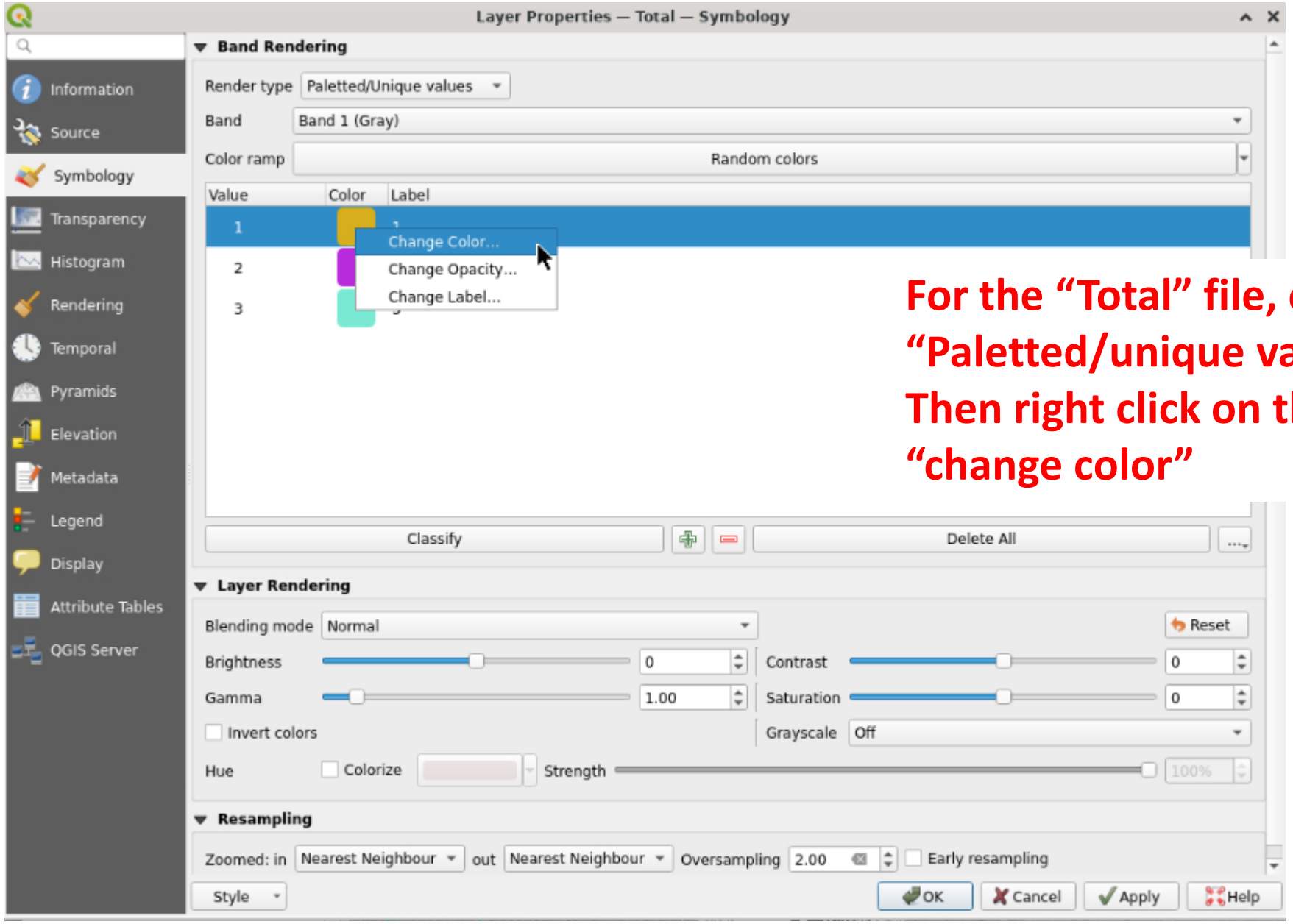




- LAHARZ made a bunch of different files
- There is a separate file for each volume/initiation point combination (these start with “IP”
- LAHARZ also made a file for each volume, merging all the flows from each initiation point (these start with “V”)
- And finally, there is a “Total” file that has all the volumes from all the initiation points in one file
- Add the “total.tif” file to your map

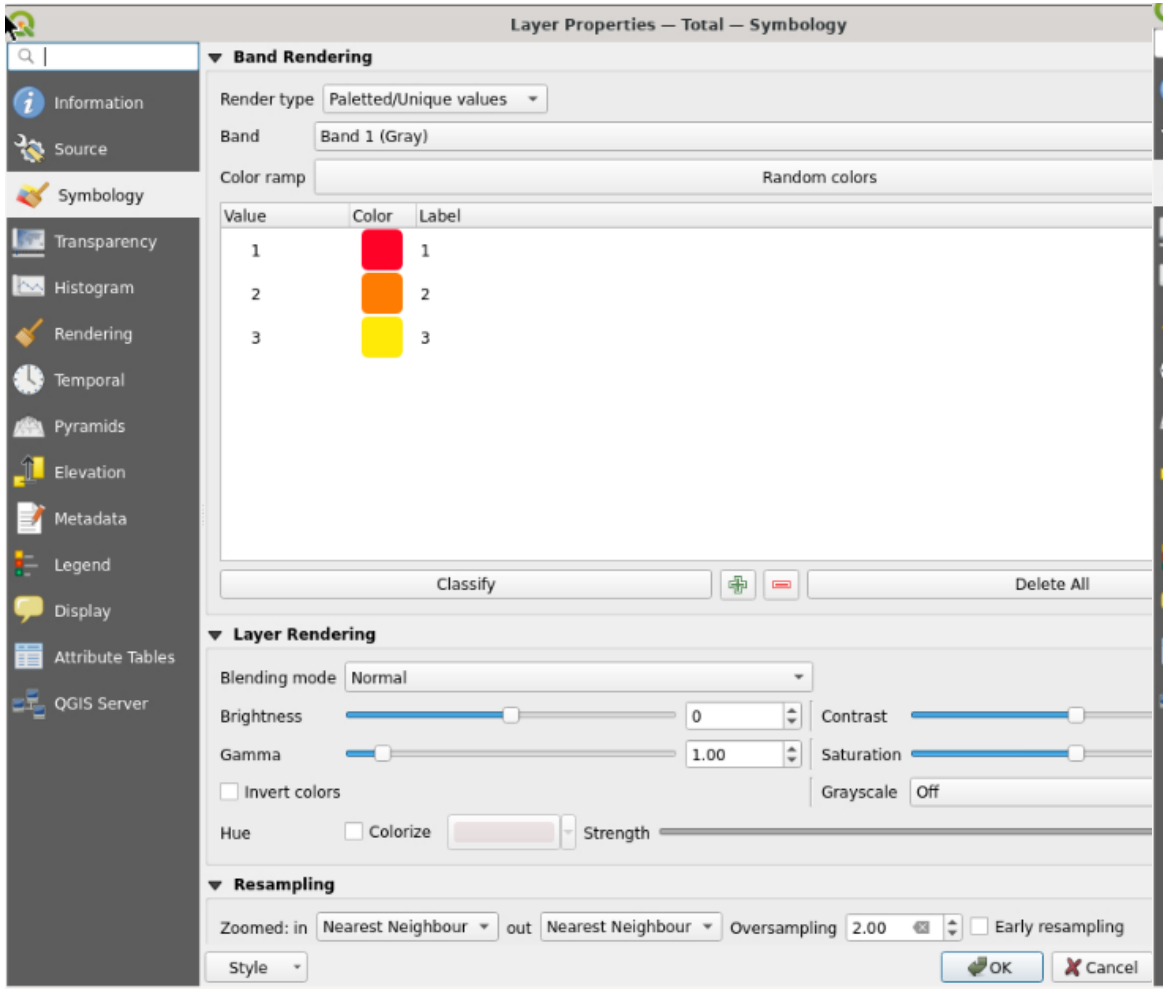
Let’s change the colors and make the map prettier.

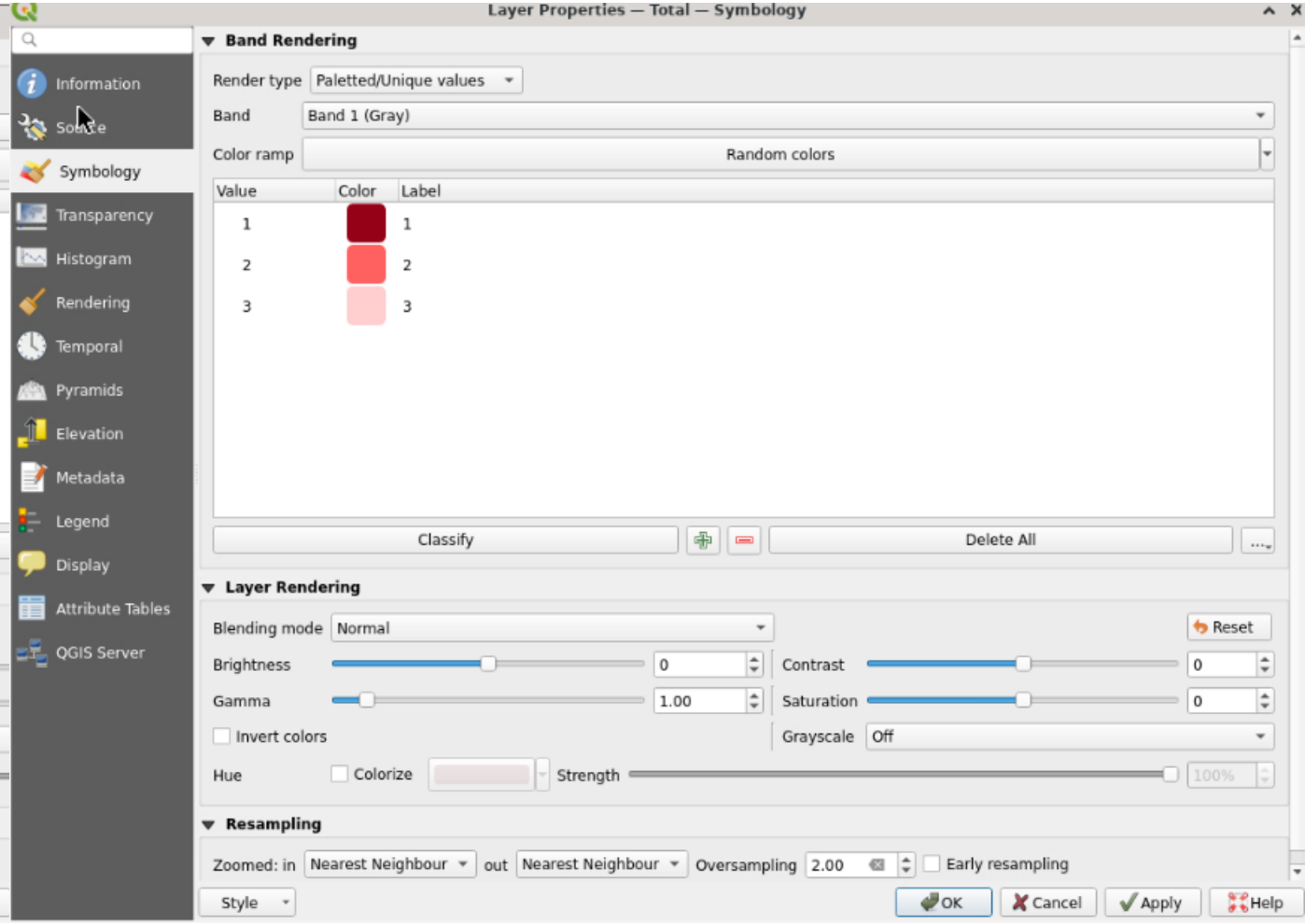


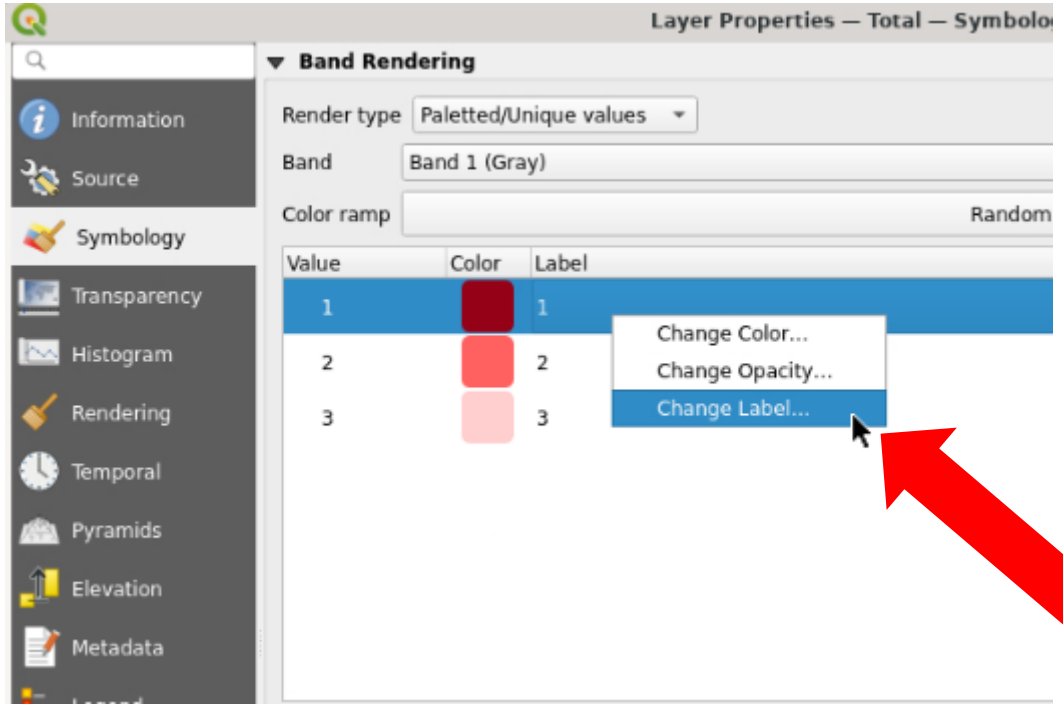


For the “Total” file, change the “render type” to “Paletted/unique values” and click “classify”. Then right click on the listed value and choose “change color”

- The lowest value (1) is the lowest volume (1 x 10^7).
- This is the highest hazard zone because smaller lahars occur more often and even a small lahar could reach this area.
- Choose a color scheme that makes sense, like red-orange-yellow or dark red to light red.
- We'll talk more about color schemes later!

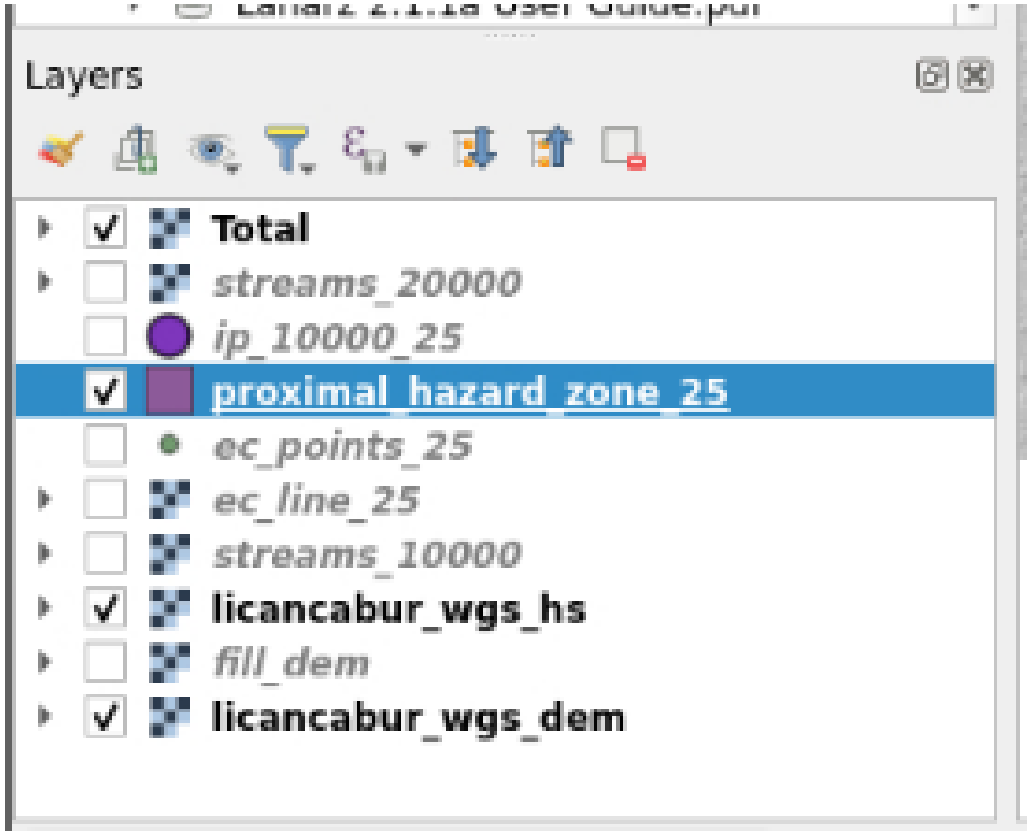




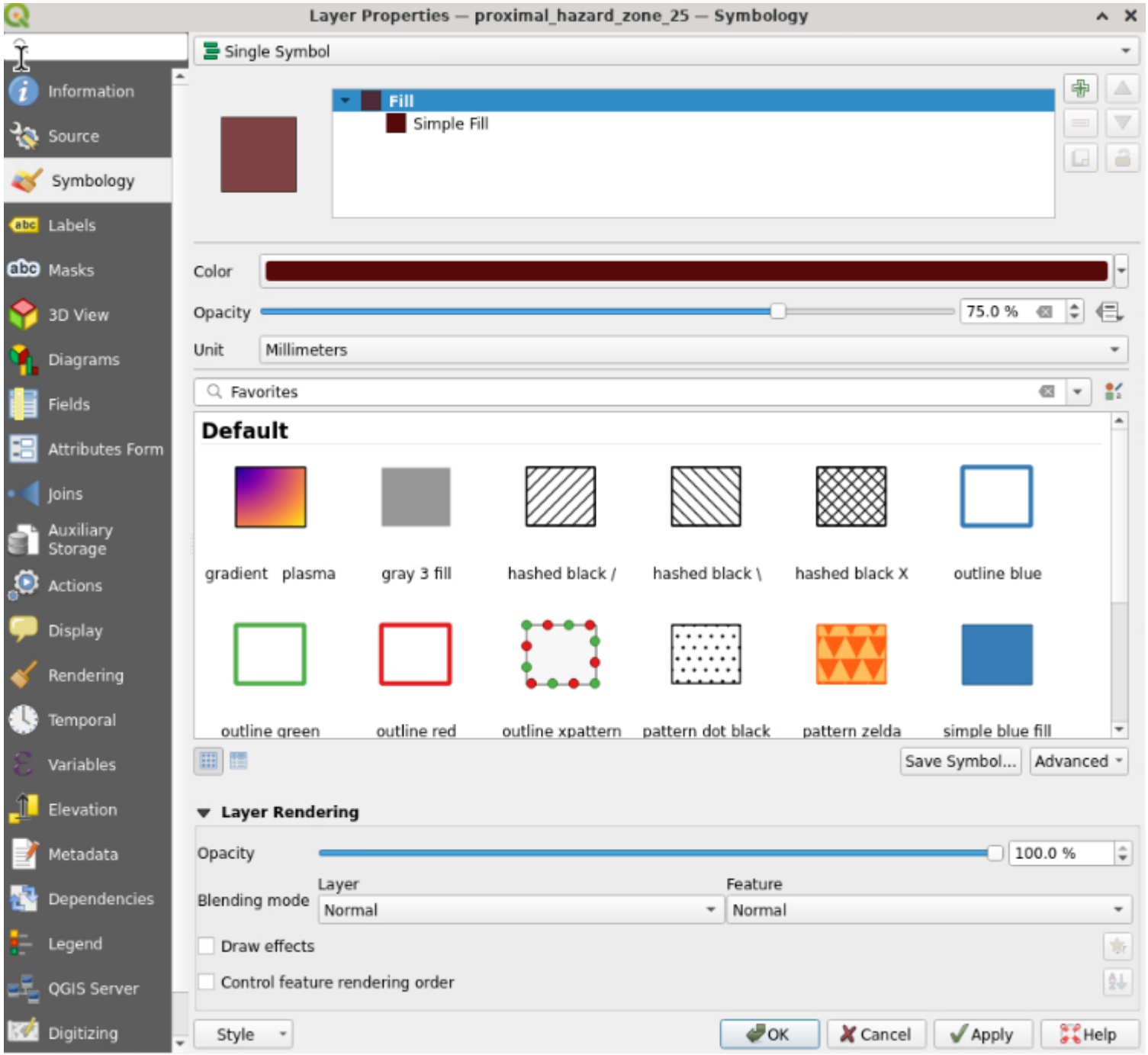


You can also change the label for each layer to: 0.01 km3, 0.03 km3, 0.1 km3

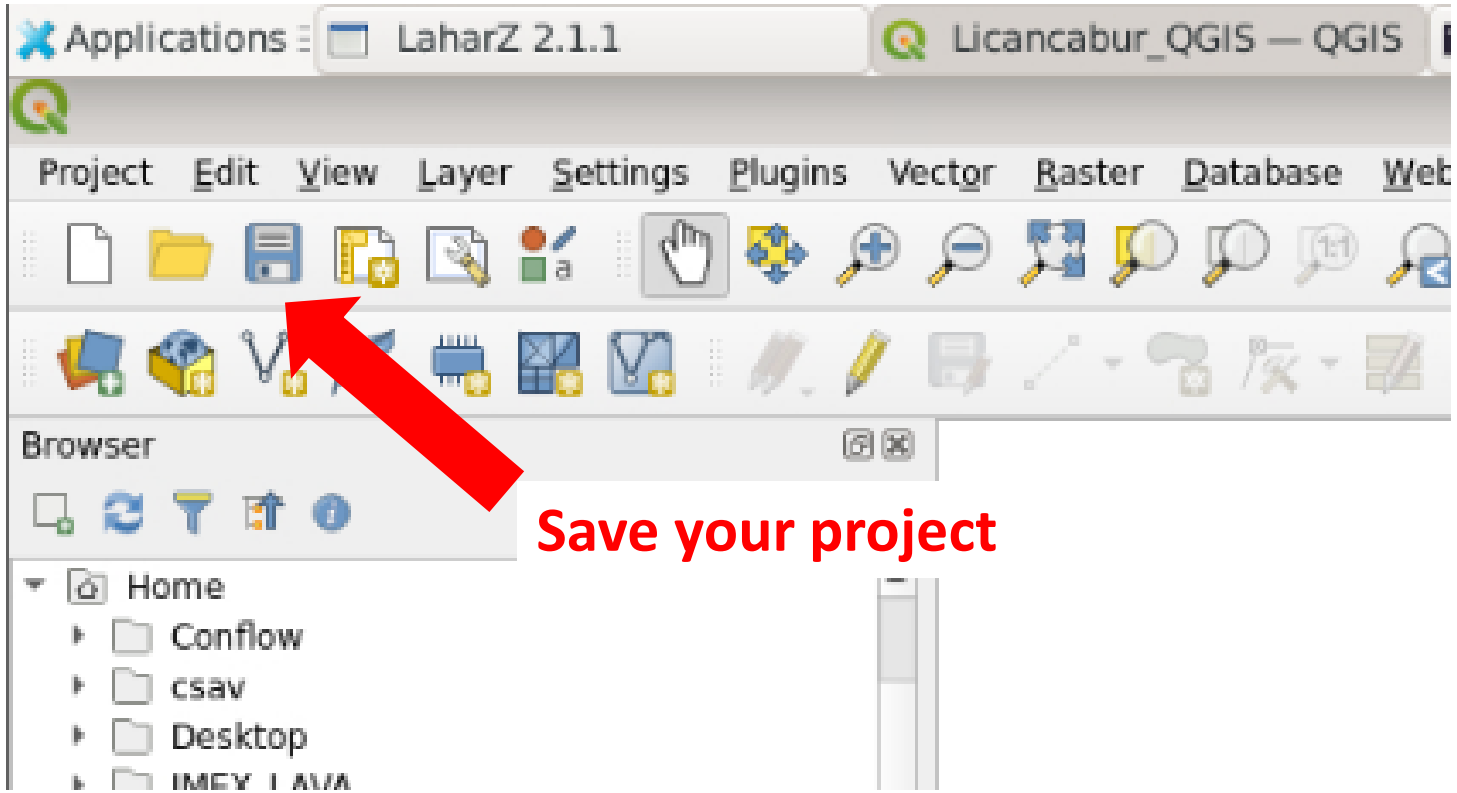
Let’s make the map neater



- Turn off layers that aren’t needed



Change the color and opacity of the proximal hazard zone layer


Applications
LaharZ 2.1.1
Licancabur_QGIS — QGIS
Project
Edit
View
Layer
Settings
Plugins
Vector
Raster
Database
Web
Browser
Save your project
Home
Conflow
csav
Desktop

Congratulations!

¡Felicidades!

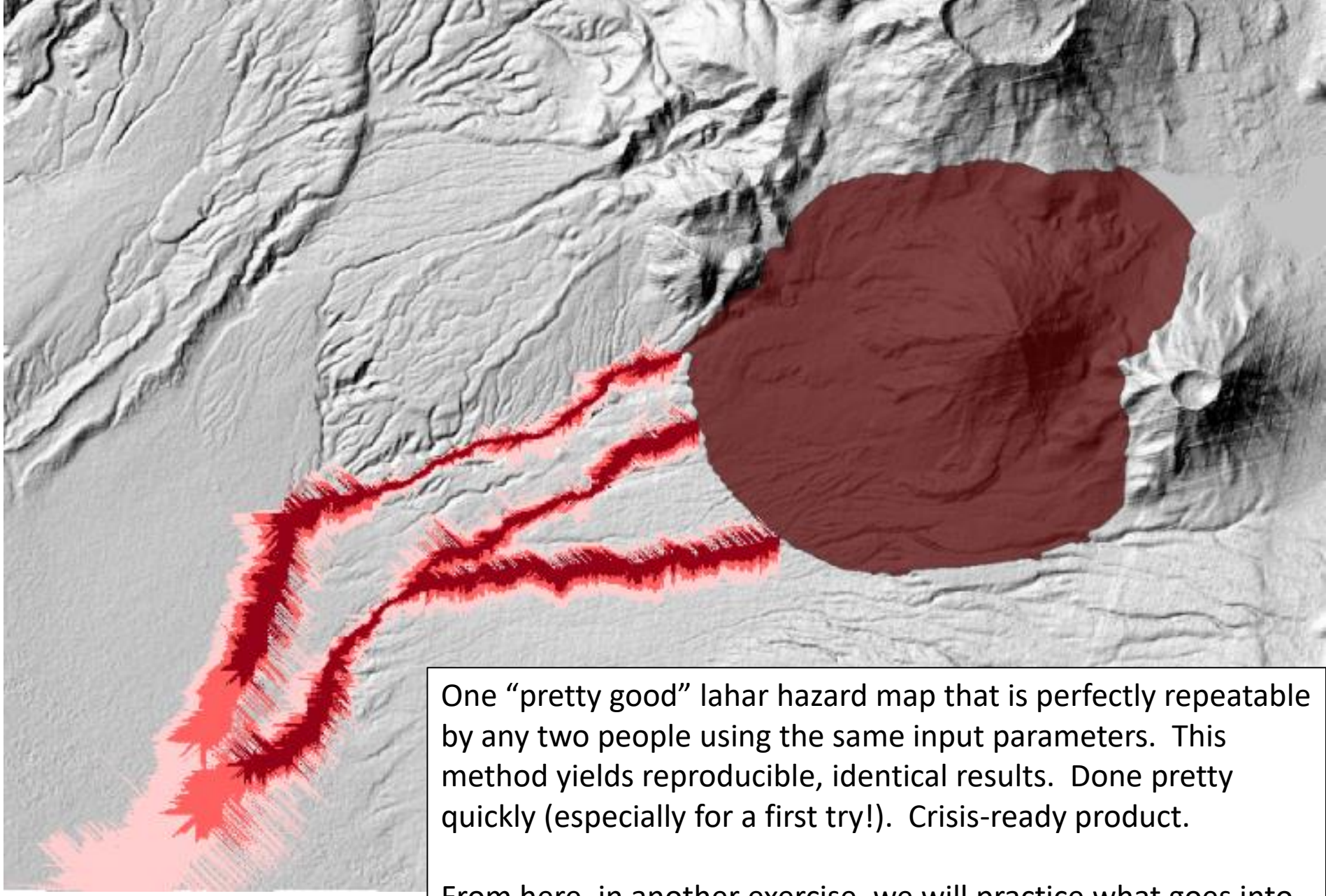
تهانينا

Selamat!

Binabati kita!

Baie geluk!

Félicitations!



One “pretty good” lahar hazard map that is perfectly repeatable by any two people using the same input parameters. This method yields reproducible, identical results. Done pretty quickly (especially for a first try!). Crisis-ready product.

From here, in another exercise, we will practice what goes into the decisions of the H/L cone, volumes, stream thresholds, and map design.